Introduction to Deep Learning

sjpark@rcv.sejong.ac.kr

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1. What is Neural Network

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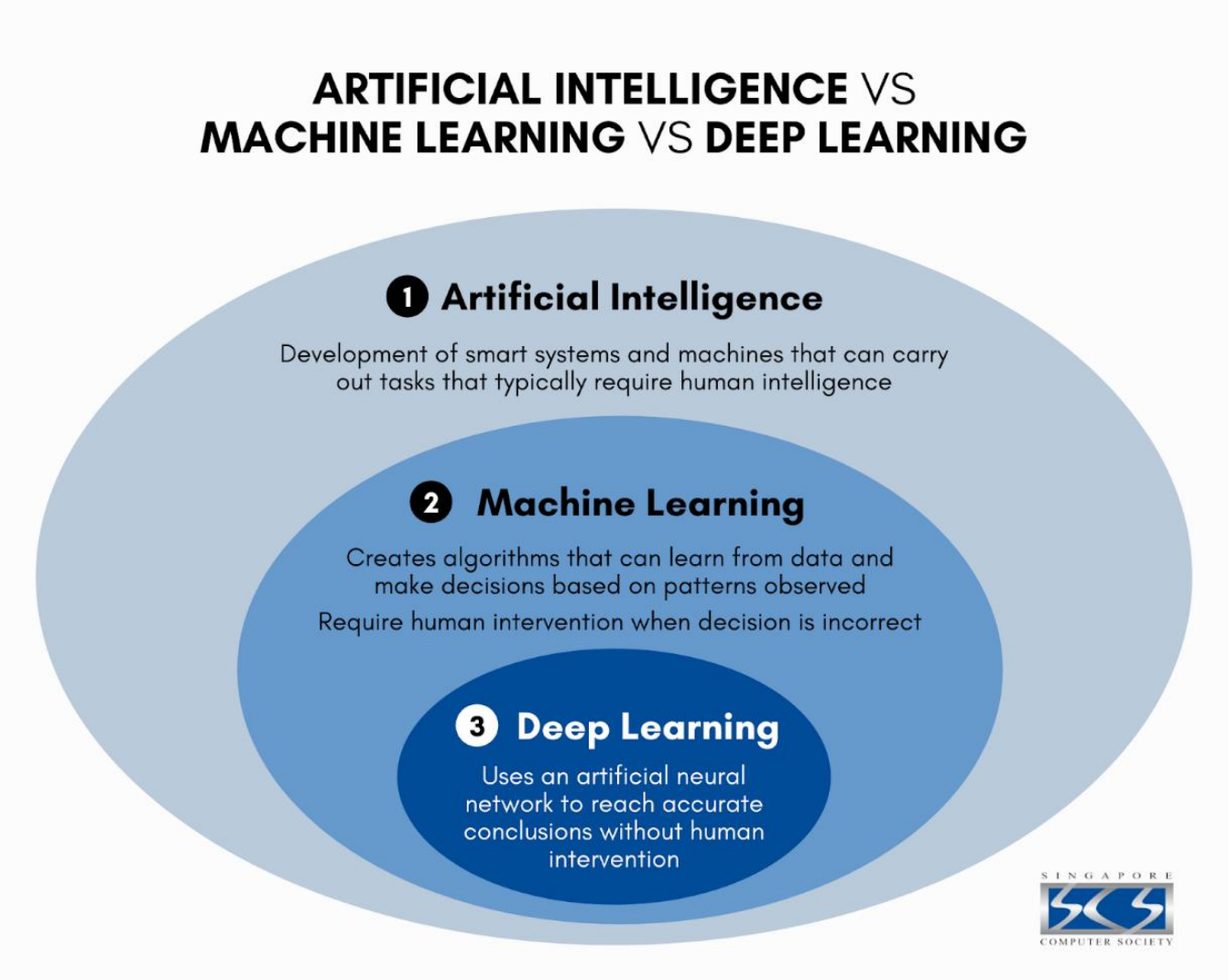
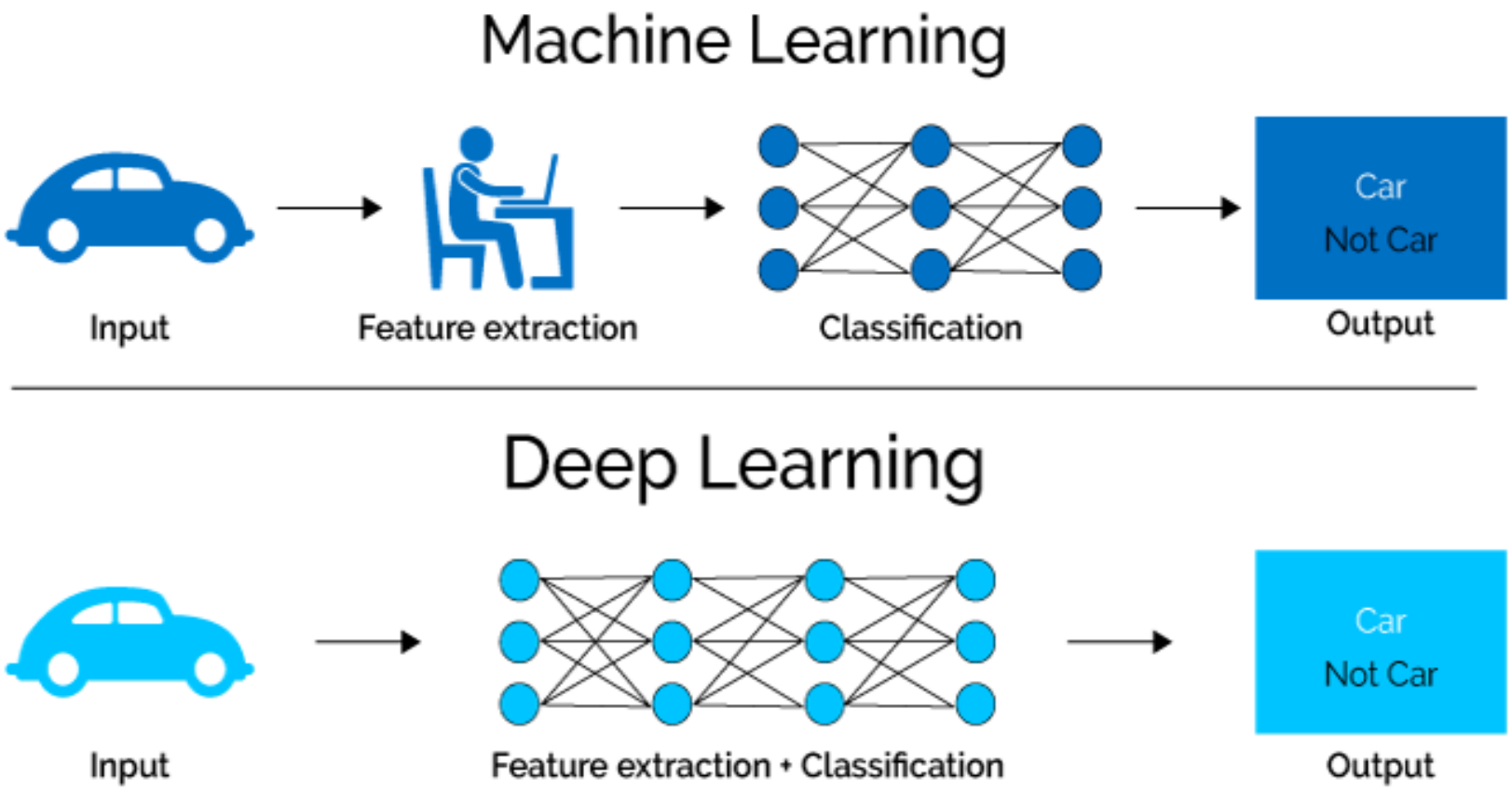
1-1. Machine Learning vs Deep Learning

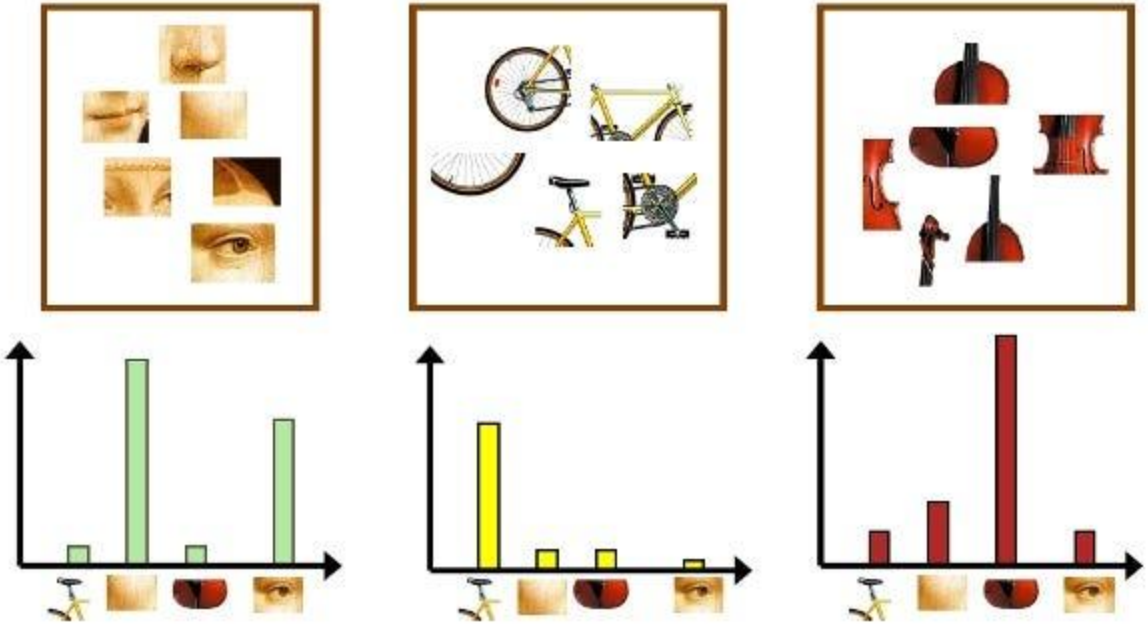


1-1. Machine Learning vs Deep Learning



1-1. Machine Learning vs Deep Learning

Example of Machine Learning: BoVW



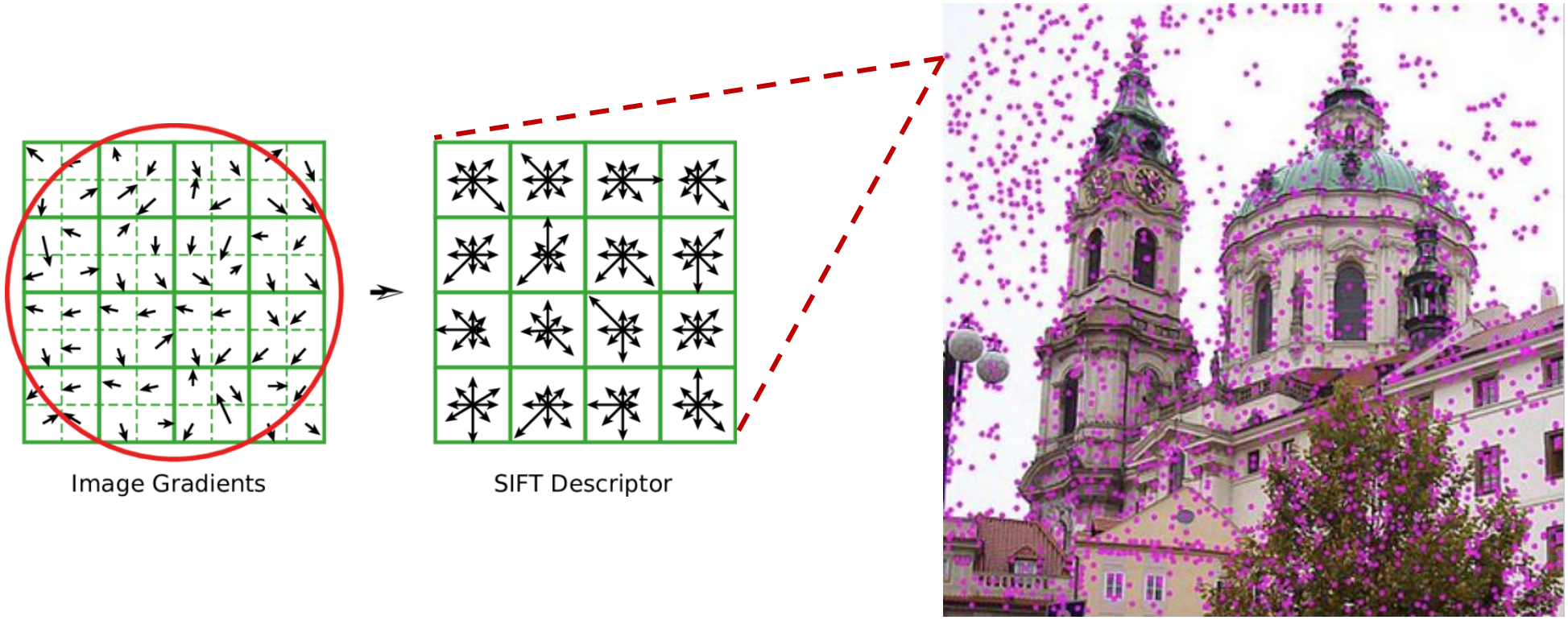
<Bag of Visual Word (BoVW)>

A definition of the BoVW model can be the
"**histogram representation based on independent features**"

1-1. Machine Learning vs Deep Learning

Example of Machine Learning: BoVW

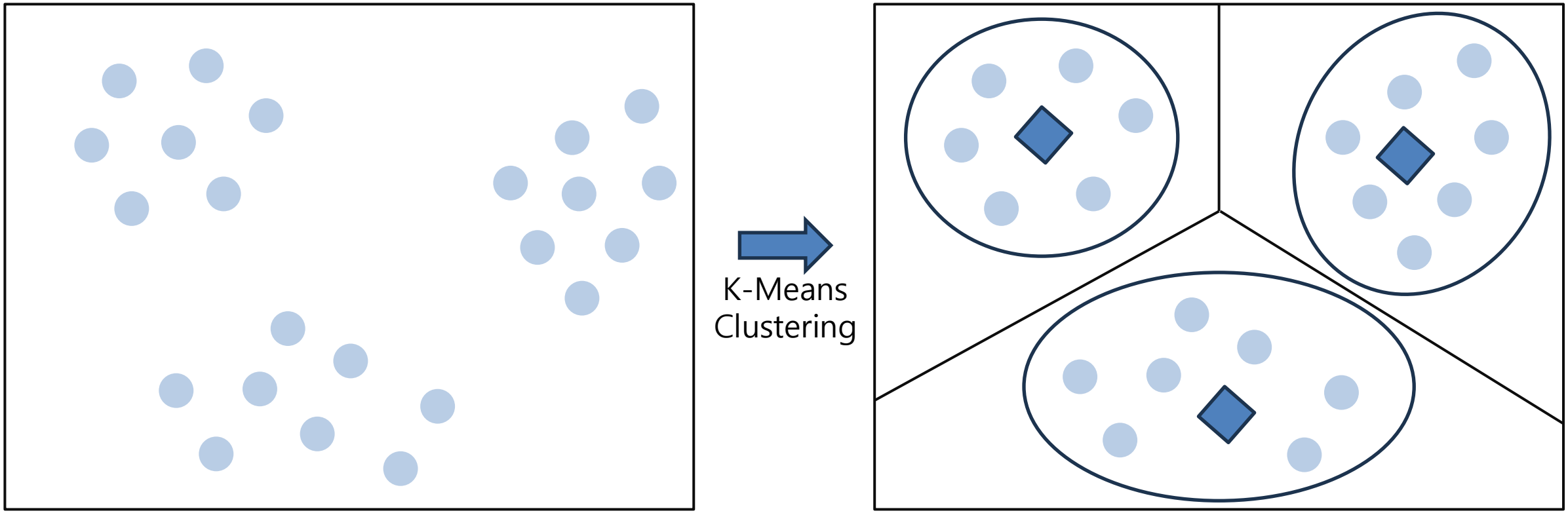
SIFT : Scale-Invariant Feature Transform



Step 1) Aggregate the **<u>SIFT descriptors</u>** of all the images

1-1. Machine Learning vs Deep Learning

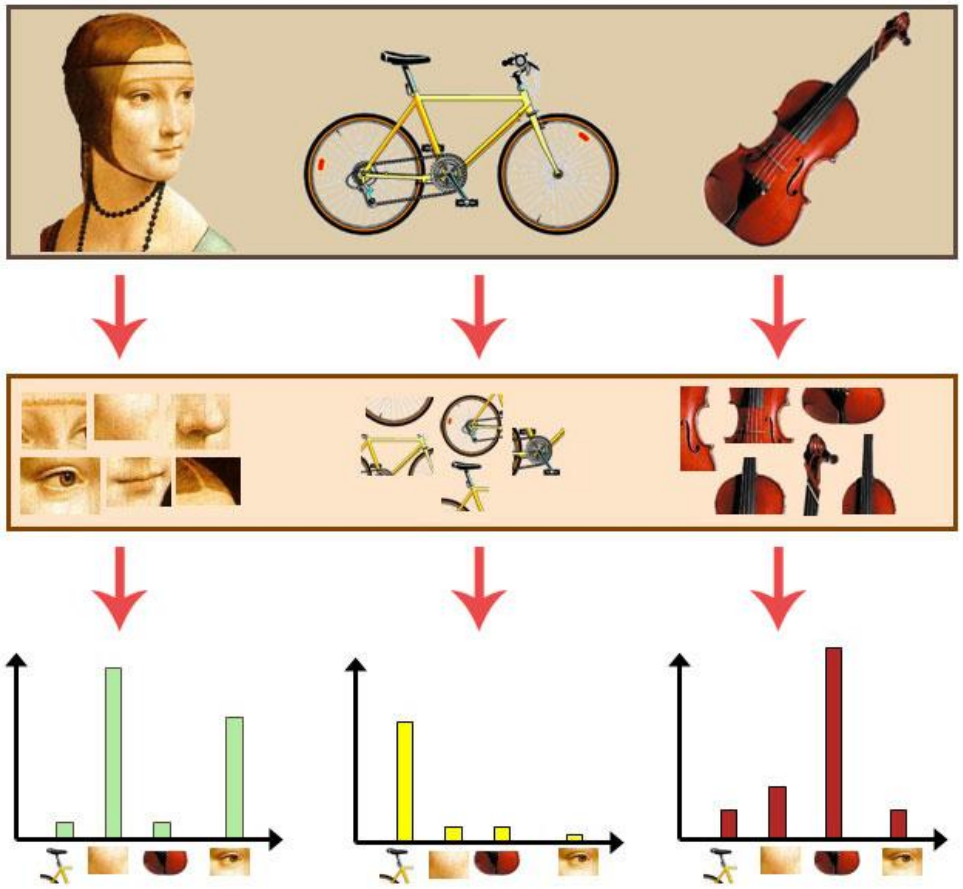
Example of Machine Learning: BoVW



Step 2) Generate Codebook via Clustering

1-1. Machine Learning vs Deep Learning

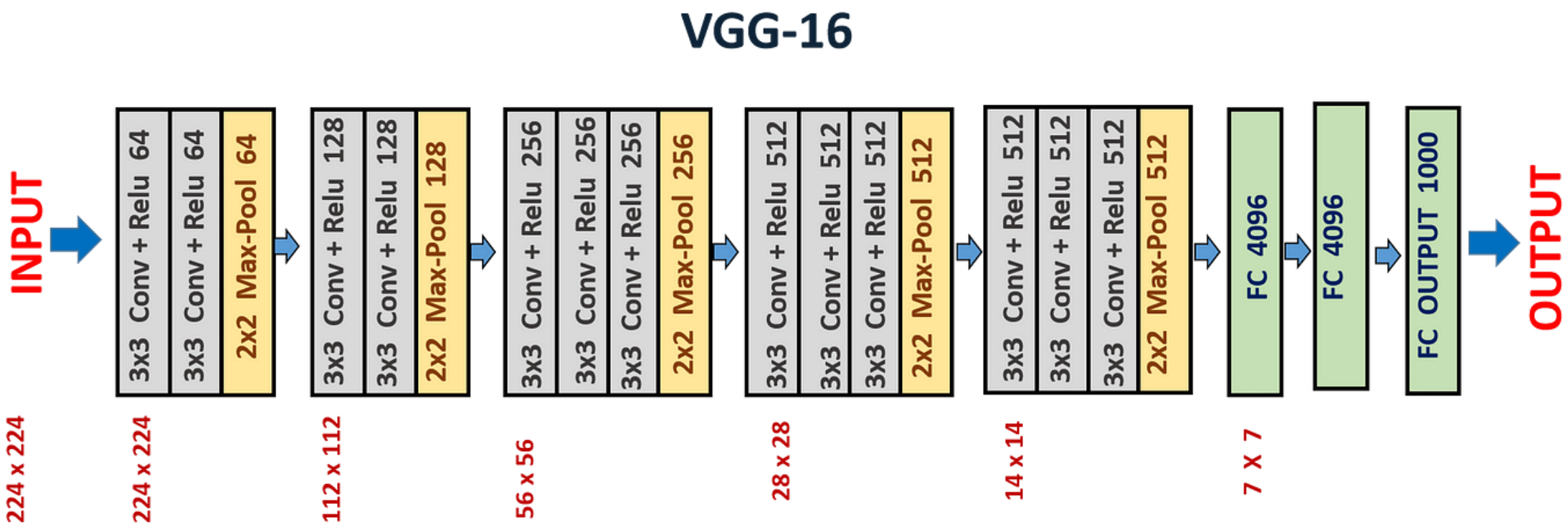
Example of Machine Learning: BoVW



Step 3) Represent image as histogram via codeword

1-1. Machine Learning vs Deep Learning

Example of Deep Learning: VGG16



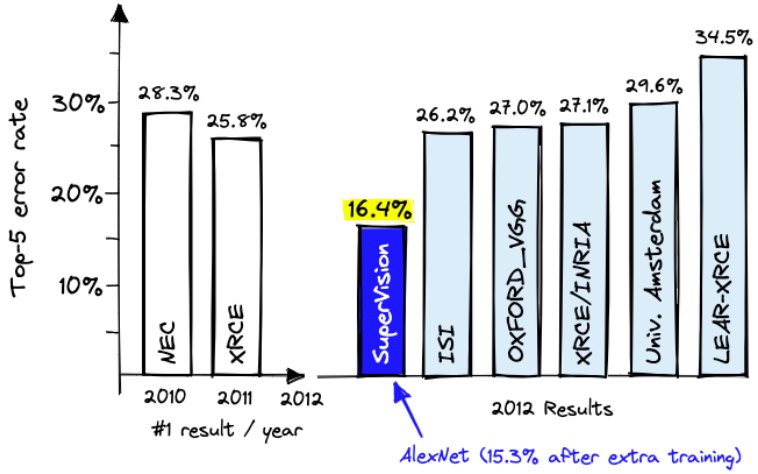
1-1. Machine Learning vs Deep Learning

ImageNet Large Scale Visual Recognition Challenges (ILSVRC)

ImageNet: 14 million images classified by WordNet hierarchy of nouns

1-1. Machine Learning vs Deep Learning

Example of Deep Learning: AlexNet



ILSVRC 2012

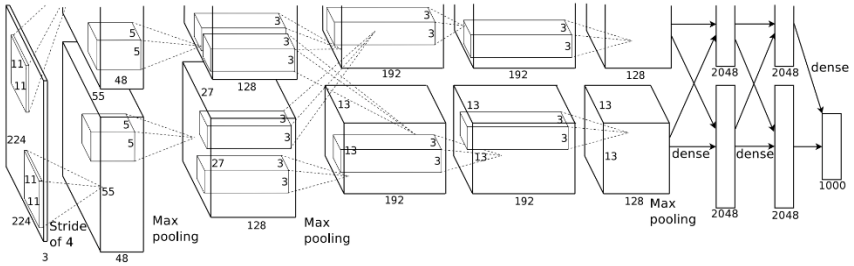


Figure 2: An illustration of the architecture of our CNN, explicitly showing the delineation of responsibilities between the two GPUs. One GPU runs the layer-parts at the top of the figure while the other runs the layer-parts at the bottom. The GPUs communicate only at certain layers. The network's input is 150,528-dimensional, and the number of neurons in the network's remaining layers is given by 253,440–186,624–64,896–64,896–43,264–4096–4096–1000.

AlexNet[1]

Task 1

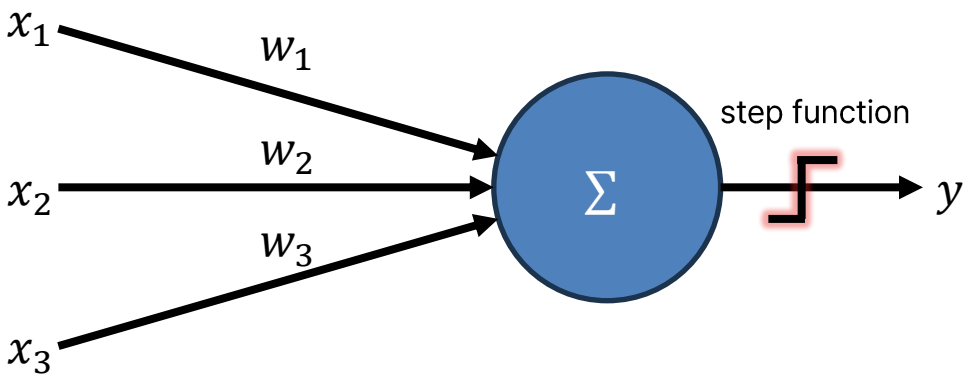
Team name	Filename	Error (5 guesses)	Description
SuperVision	test-preds-141-146.2009-131-137-145-146.2011-145f.	0.15315	Using extra training data from ImageNet Fall 2011 release
SuperVision	test-preds-131-137-145-135-145f.txt	0.16422	Using only supplied training data
ISI	pred_FVs_wLACs_weighted.txt	0.26172	Weighted sum of scores from each classifier with SIFT+FV, LBP+FV, GIST+FV, and CSIFT+FV, respectively.
ISI	pred_FVs_weighted.txt	0.26602	Weighted sum of scores from classifiers using each FV.

[1] ImageNet Classification with Deep Convolutional Neural Networks, Alex Krizhevsky, Ilya Sutskever, Geoffrey E. Hinton (2012, NIPS)

1-2. Neural Network

Perceptron (Linear Model)

- Type of **<u>linear classifier</u>**, i.e. a classification algorithm that makes its predictions based on a <u>linear predictor function combining a set of weights with the feature vector</u>

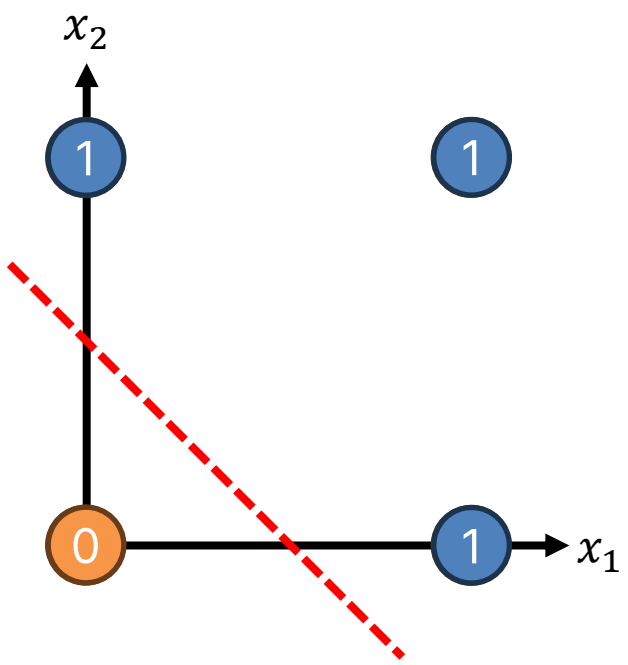


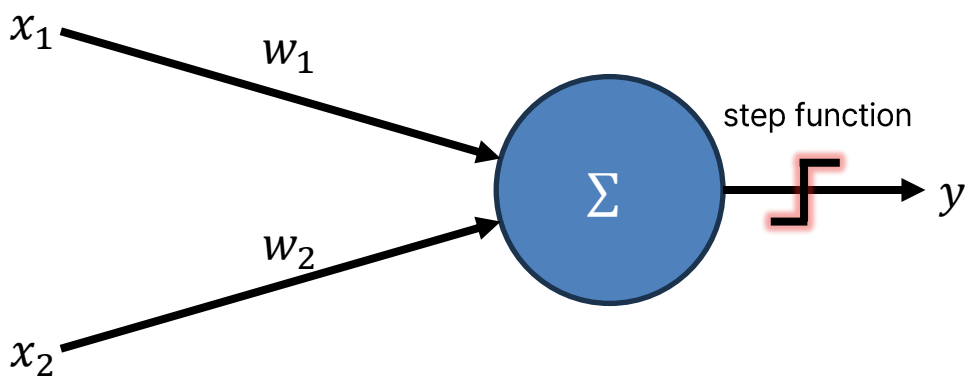
$$y = s\left(\sum_{i=0}^{N} w_i x_i\right)$$

$$s(v) = \begin{cases} 1 & \text{if } v > \theta \\ 0 & \text{otherwise} \end{cases}$$

1-2. Neural Network

Perceptron (Linear Model)





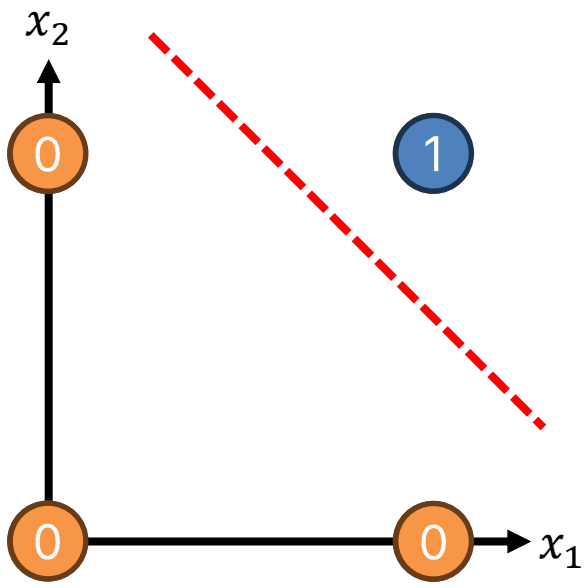
$$y = s\left(\sum_{i=0}^{N} w_i x_i\right)$$

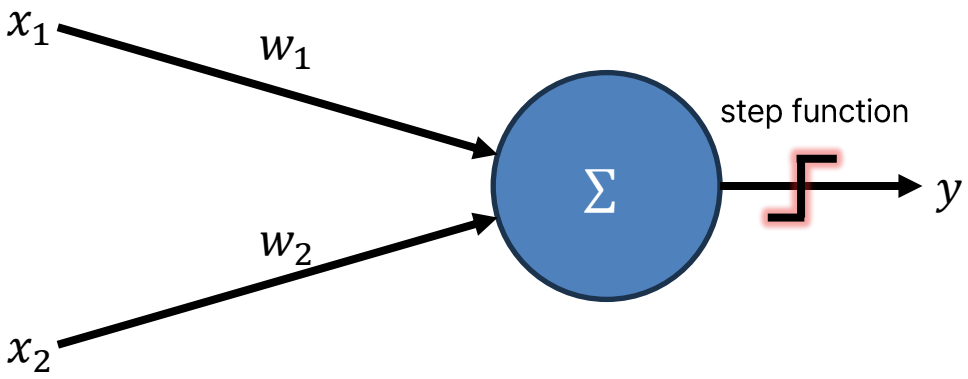
$$s(v) = \begin{cases} 1 & \text{if } v > 0.6 \\ 0 & \text{otherwise} \end{cases}$$

Let $w_1 = 0.7$ and $w_2 = 0.7$ … Perceptron works as OR gate!

1-2. Neural Network

Perceptron (Linear Model)





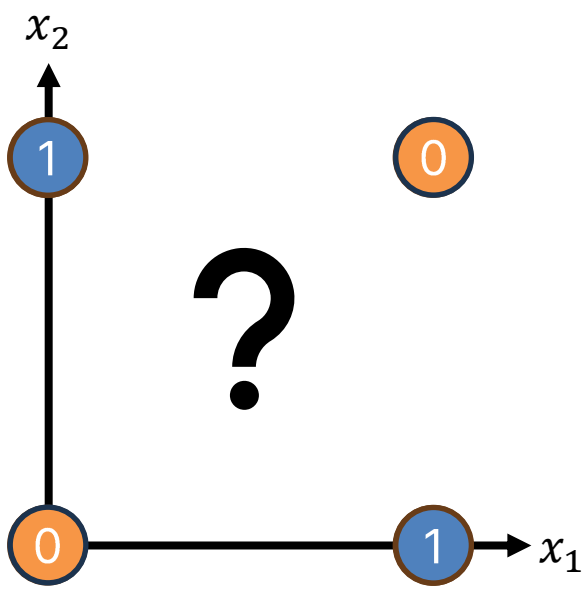
$$y = s\left(\sum_{i=0}^{N} w_i x_i\right)$$

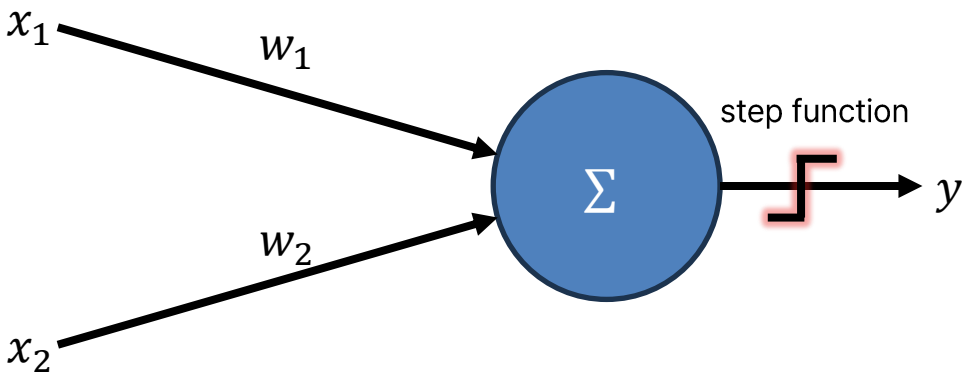
$$s(v) = \begin{cases} 1 & \text{if } v > 0.6 \\ 0 & \text{otherwise} \end{cases}$$

When $w_1 = 0.5$ and $w_2 = 0.5$ … It works as AND gate!

1-2. Neural Network

Perceptron (Linear Model)





$$y = s\left(\sum_{i=0}^{N} w_i x_i\right)$$

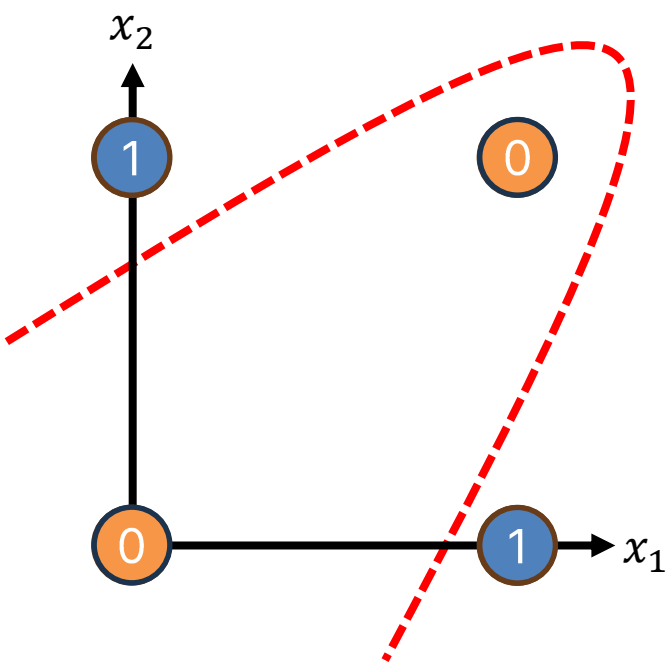
$$s(v) = \begin{cases} 1 & \text{if } v > 0.6 \\ 0 & \text{otherwise} \end{cases}$$

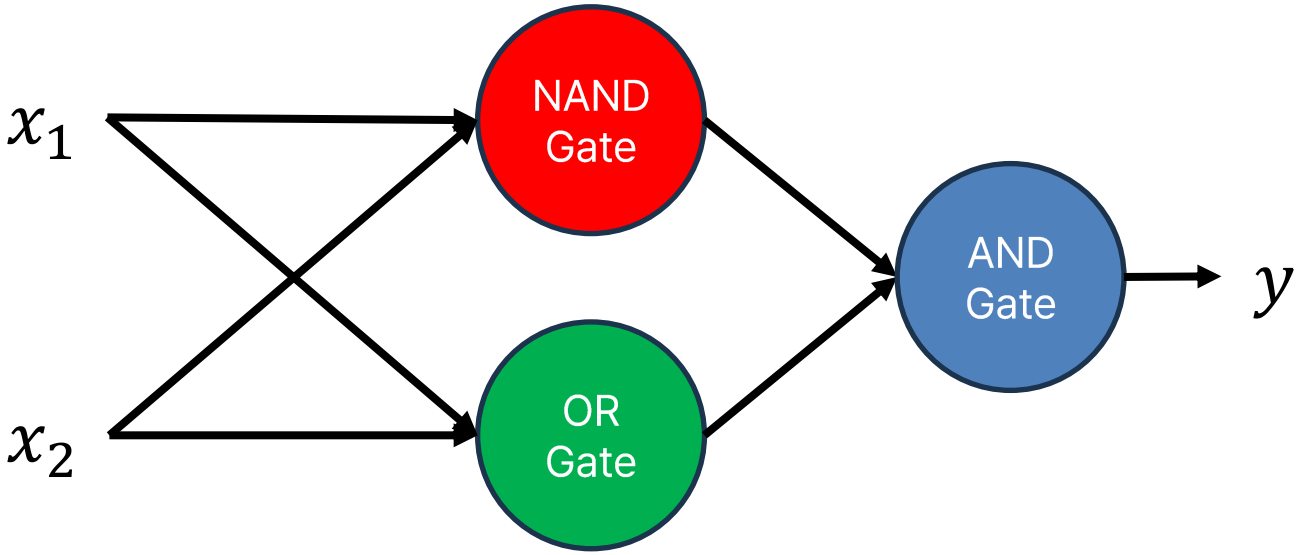
However, **<u>XOR</u>** could not be solved by a perceptron

1-2. Neural Network

Multi-Layer Perceptron (MLP)

- If the problem is not **<u>linearly separable</u>**, a single perceptron has its limitations

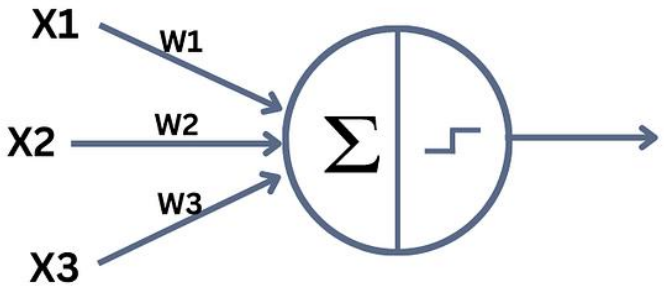




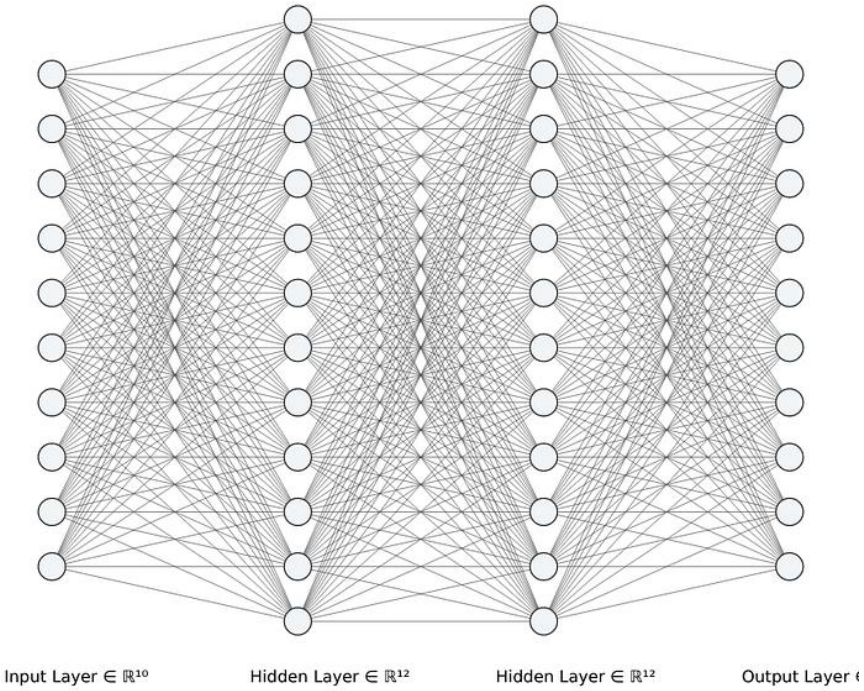
1-2. Neural Network

Multi-Layer Perceptron (MLP)

- With the **<u>continuous combination of linear units and non-linear functions,</u>** neural networks have begun to solve complex problems



Single-layer perceptron



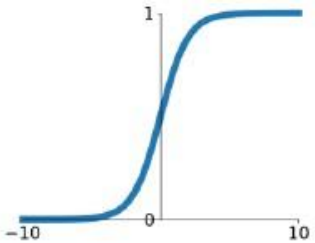
Multi-layer perceptron

1-2. Neural Network

Activation Function (Non-Linearity)

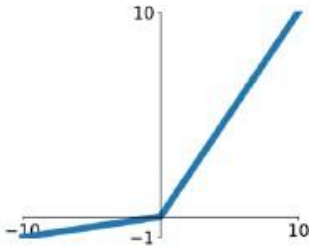
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



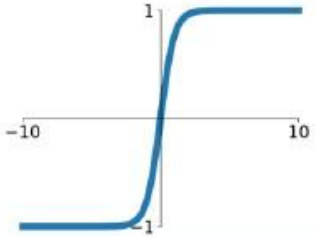
Leaky ReLU

$$\max(0.1x, x)$$



tanh

$$\tanh(x)$$

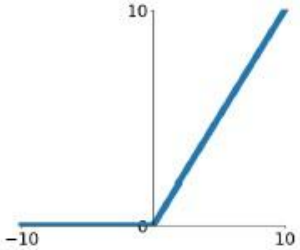


Maxout

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

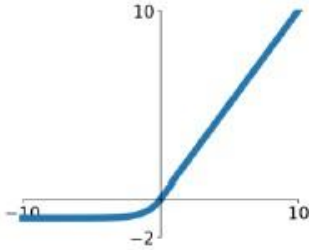
ReLU

$$\max(0, x)$$

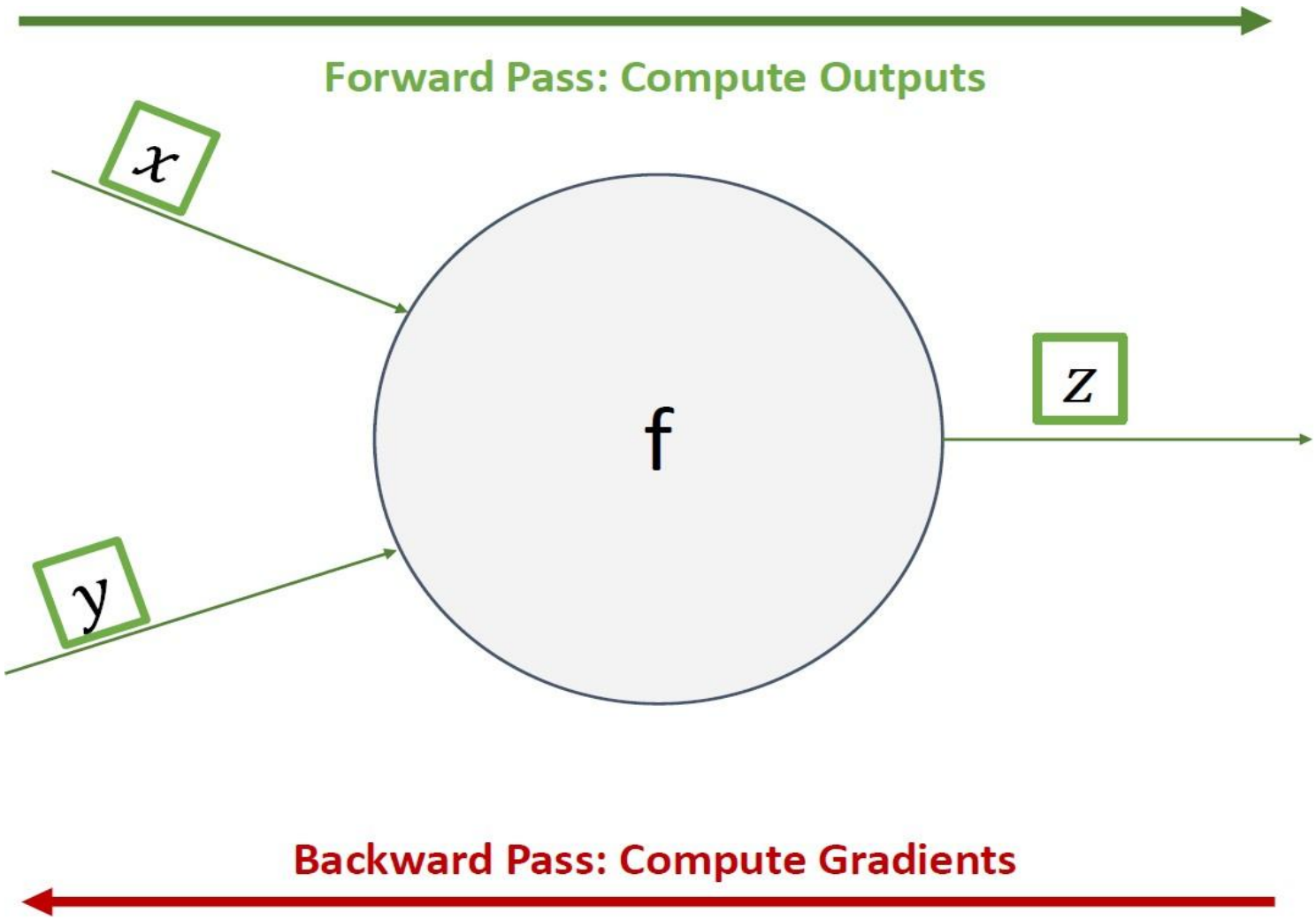


ELU

$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$



1-3. Forward / Backward Pass



1-3. Forward / Backward Pass

Training Algorithm:

1. Initialize the weight parameter w with random values
2. Process the data and make prediction
3. Calculate the error between the network's predictions and the actual labels
4. Update the parameter w in the direction that reduces the error
5. Repeat 2~4 steps until the network's predictions become accurate

1-3. Forward / Backward Pass

Loss Function

A **loss function** tells how good our current classifier is

Low loss = good classifier
High loss = bad classifier

(Also called: **objective function**; **cost function**)

Given a dataset of examples

$$\{(x_i, y_i)\}_{i=1}^{N}$$

Where x_i is an vector (image) and
y_i is an integer-valued label

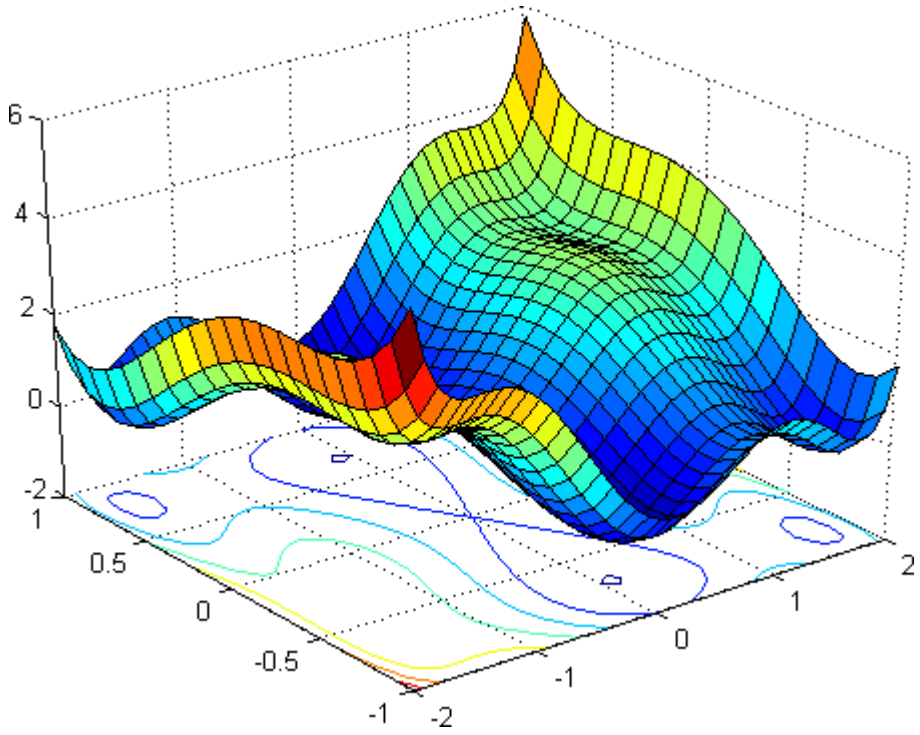
Loss for a single example is

$$L_i = L(f(x_i, W), y_i)$$

Loss for the dataset is average of per-example losses:

분류기의 Loss ➔

$$\phi(W) = \frac{1}{N}\sum_{i=1}^{N} L(f(x_i, W), y_i)$$



1-3. Forward / Backward Pass

Loss Function

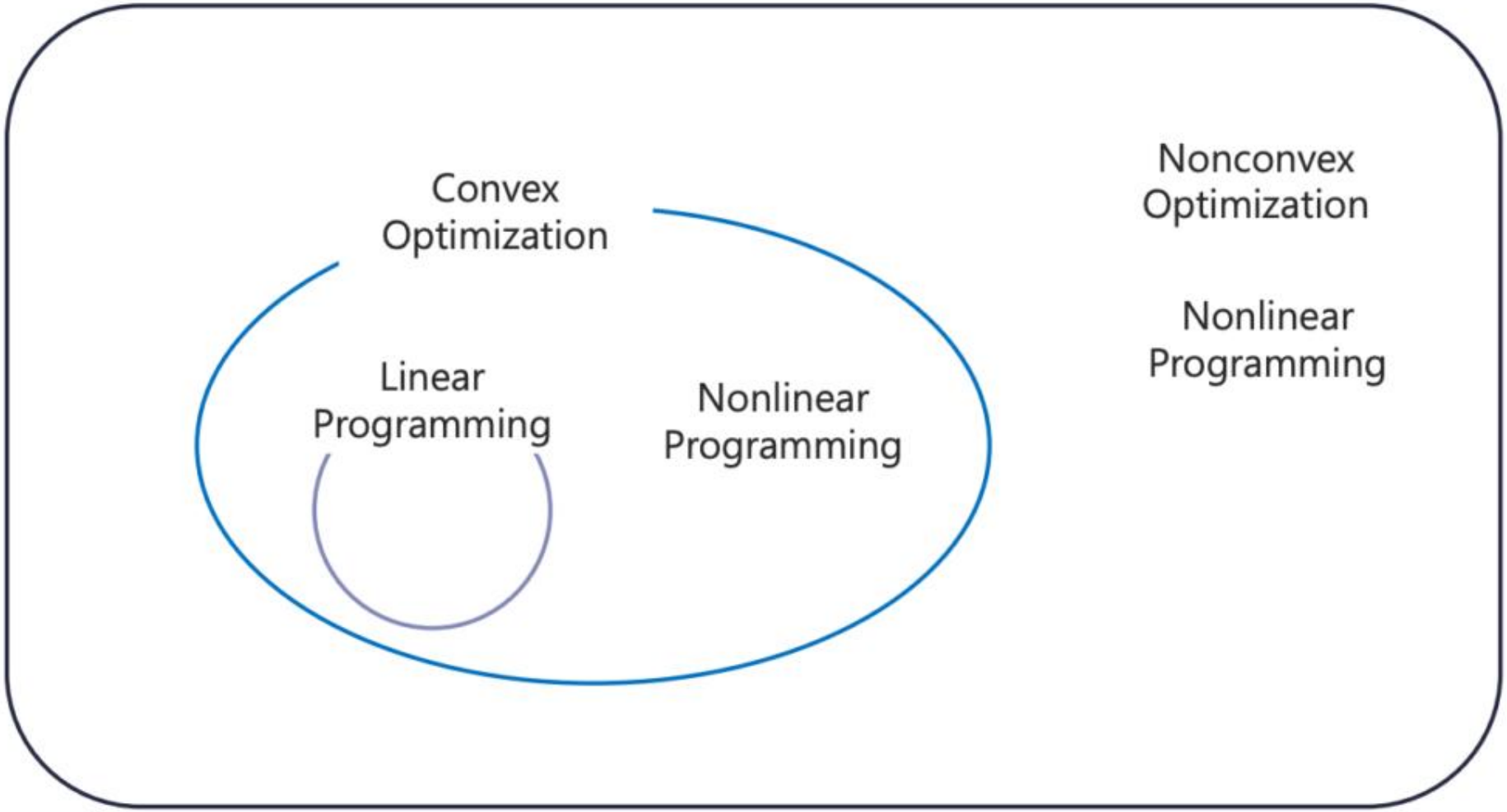
Regression Task

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}\sum_{d=1}^{D}(y_i^{(d)} - \hat{y}_i^{(d)})^2$$

Classification Task

$$\mathrm{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{d=1}^{D} y_i^{(d)} \log \hat{y}_i^{(d)}$$

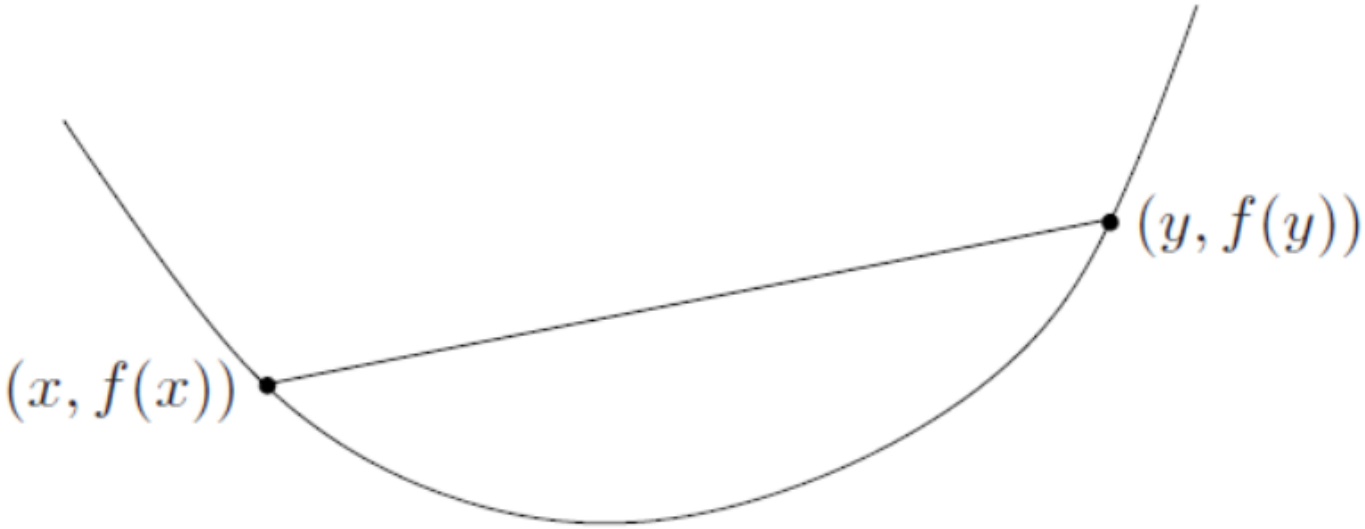
1-4. Optimization



1-4. Optimization

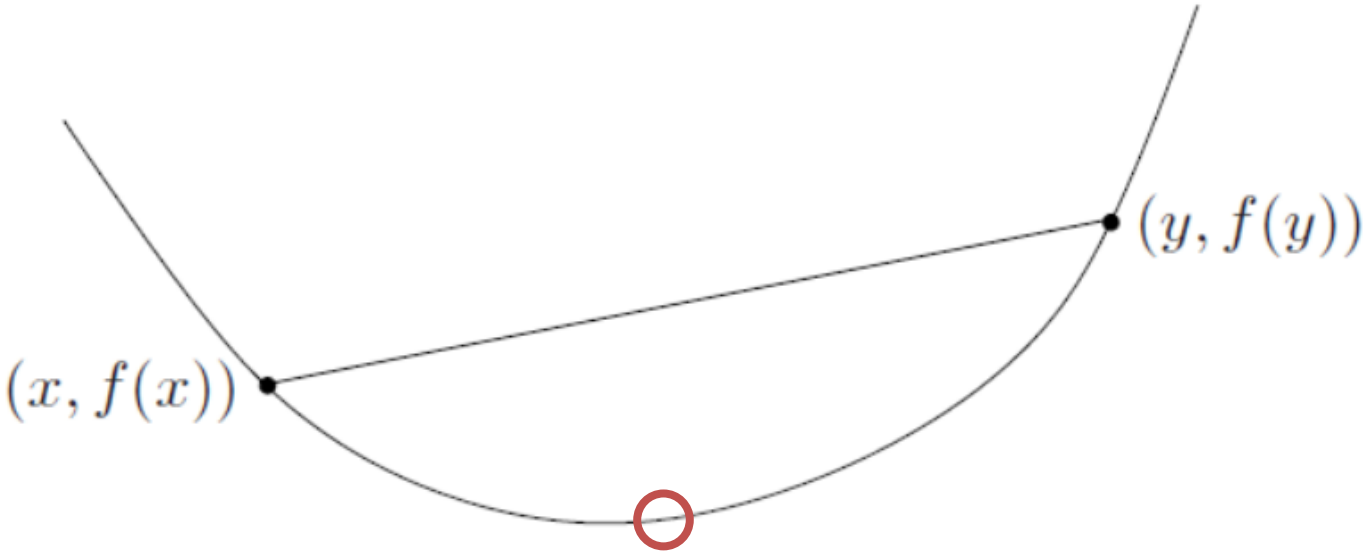
Convex function

$$f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta) f(y), 0 \leq \theta \leq 1$$



1-4. Optimization

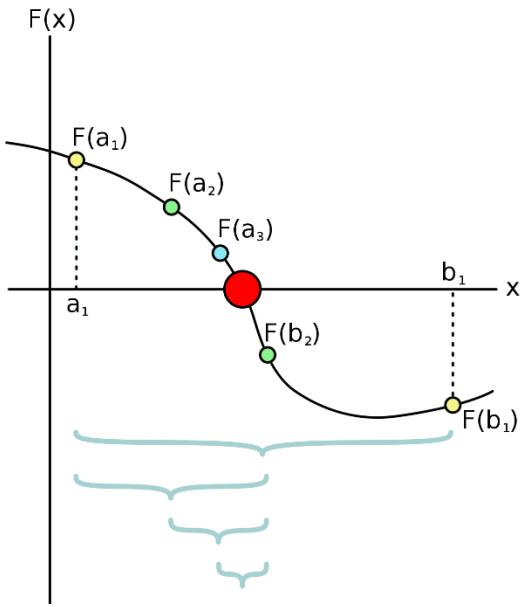
Convex function



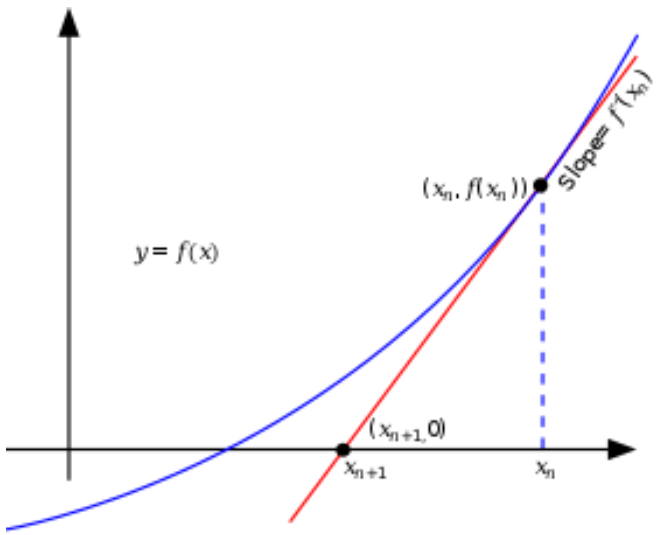
we just need to solve $\nabla f(x) = 0$

1-4. Optimization

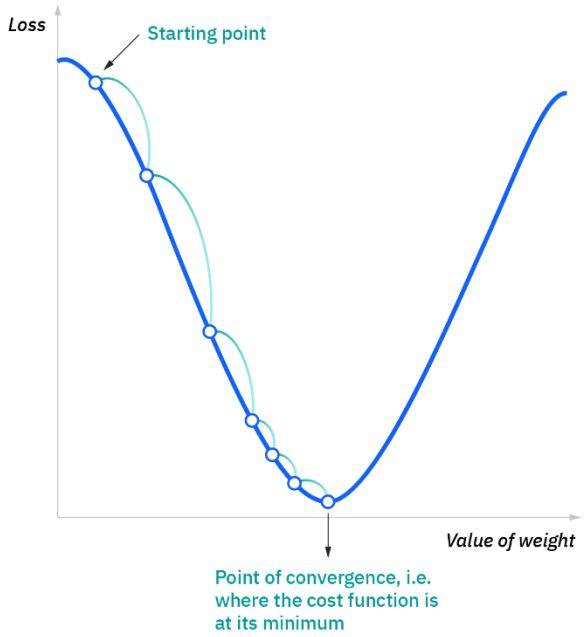
How to find minima?



Bisection method



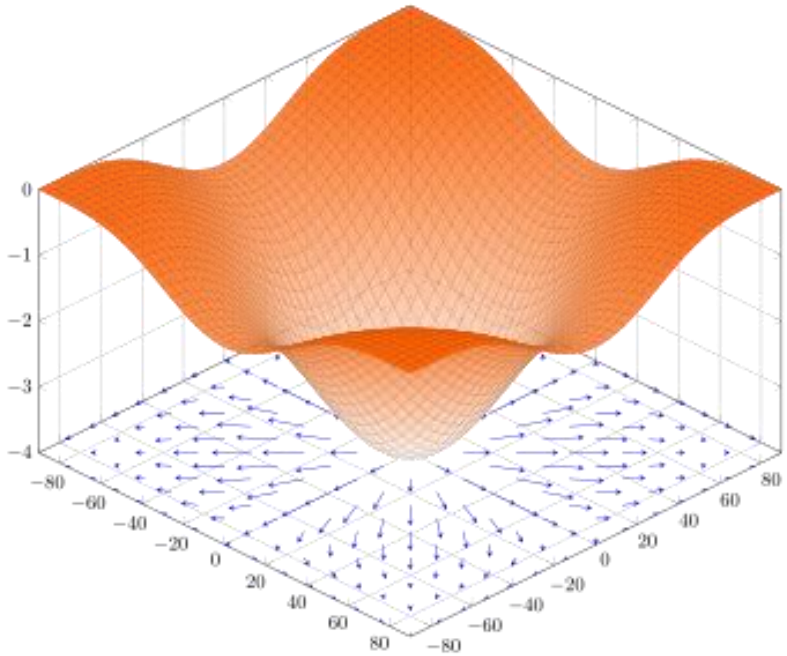
Newton's method



Gradient Descent

1-4. Optimization

What is Gradient?



$$\nabla f(p) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(p) \\ \vdots \\ \frac{\partial f}{\partial x_n}(p) \end{bmatrix}.$$

In vector calculus, the **gradient** of a scalar-valued differentiable function f of several variables is the vector field (or vector-valued function) ∇f whose value at a point p gives the direction and the rate of fastest increase

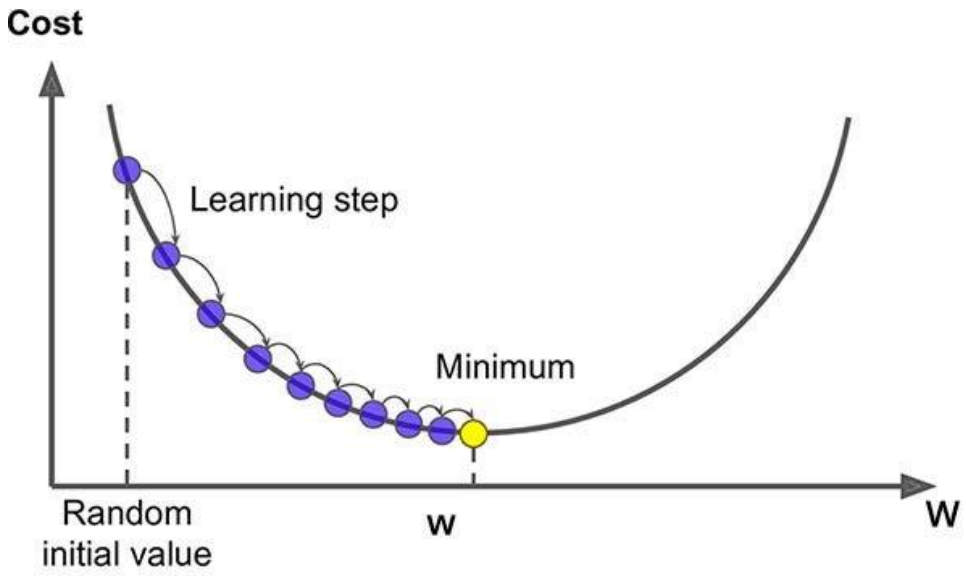
1-4. Optimization

Gradient Descent

Method: at each step, move in direction of negative gradient

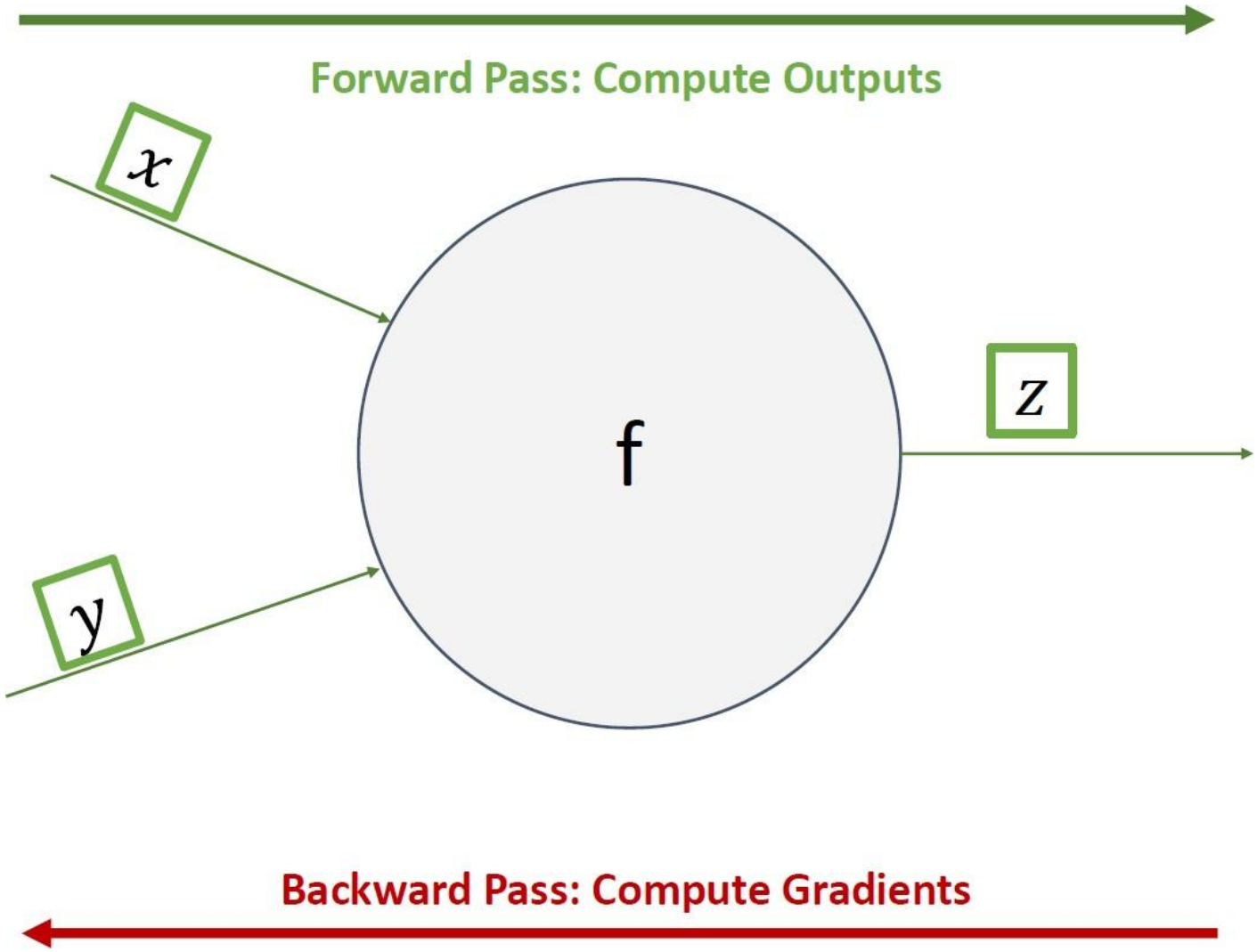
```
w0 = initialize()                    #initialize
for iter in range(numIters):
    g = ∇_w L(w)                     # eval gradient
    w = w + -stepsize*g              # update w
return w
```

Learning rate



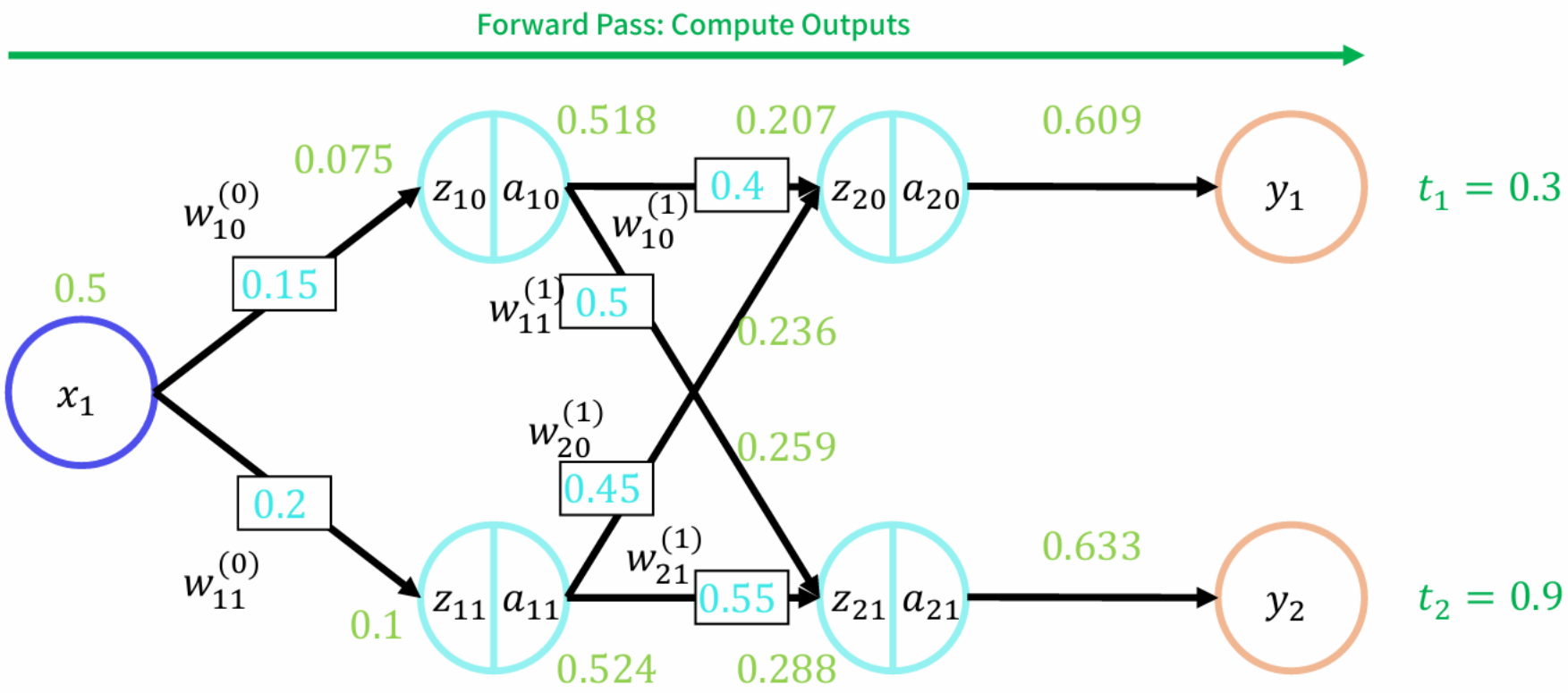
1-4. Optimization

Backpropagation



1-4. Optimization

Backpropagation

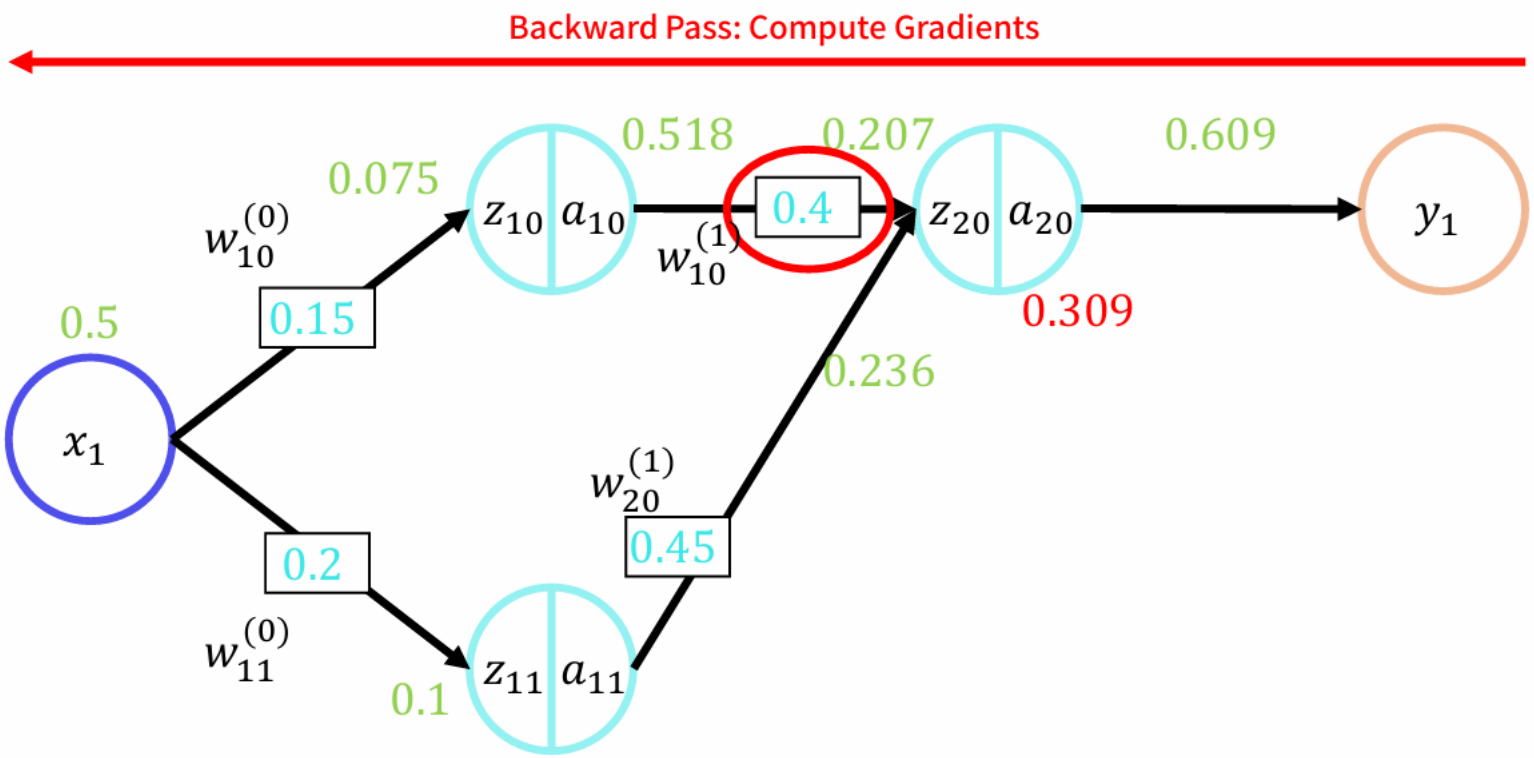


$$E = \frac{1}{2}\sum_{i=1}^{2}(t_i - y_i)^2$$

$$w_{i+1} = w - \eta * \frac{\partial E}{\partial w_i}$$

1-4. Optimization

Backpropagation



$t_1 = 0.3$

$$E = \frac{1}{2}\sum_{i=1}^{2}(t_i - y_i)^2$$

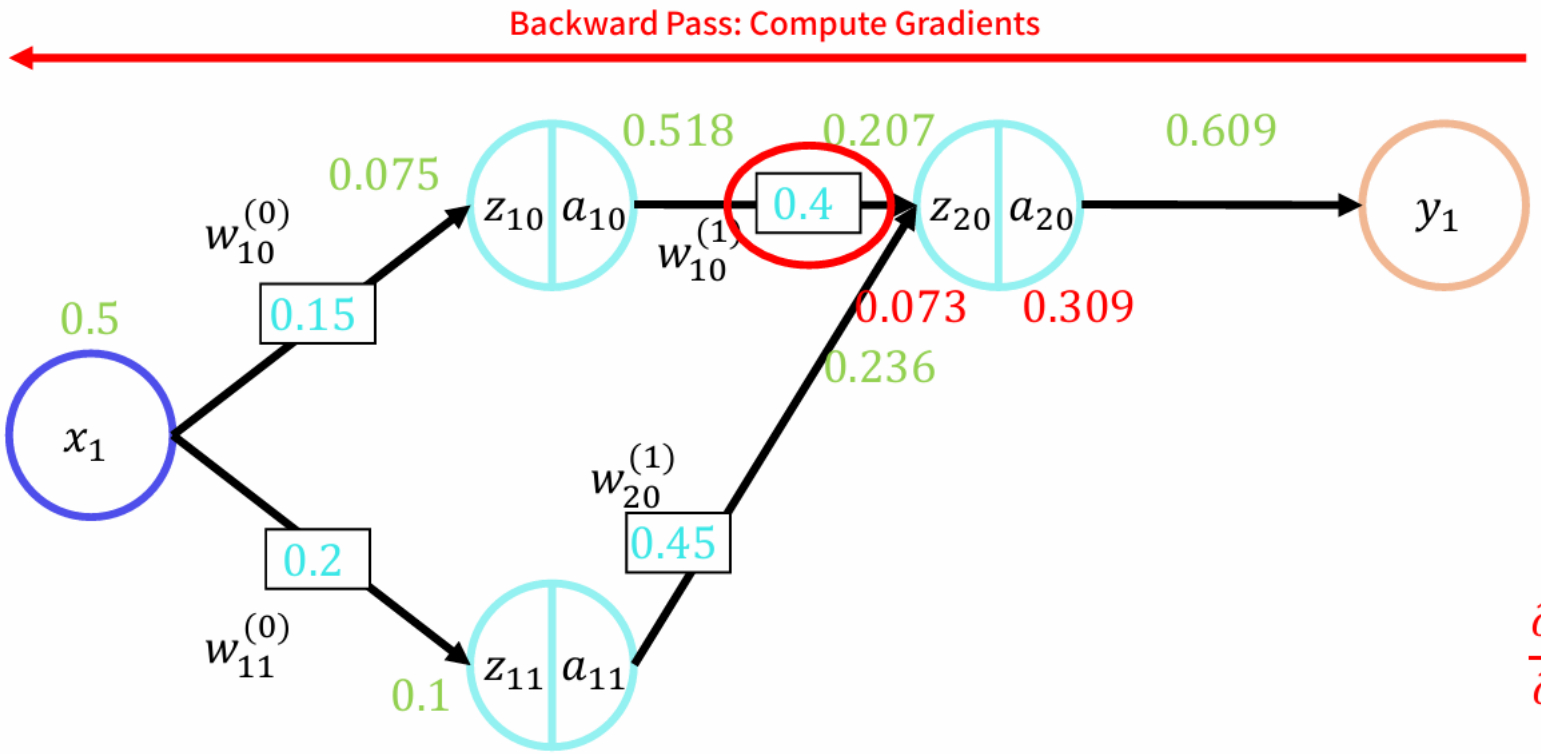
$$\frac{\partial E}{\partial w_{10}^{(1)}} = \frac{\partial E}{\partial a_{20}}\frac{\partial a_{20}}{\partial z_{20}}\frac{\partial z_{20}}{\partial w_{10}^{(1)}}$$

$$E = \frac{1}{2}\left((t_1 - a_{20})^2 + (t_2 - a_{21})^2\right)$$

$$\frac{\partial E}{\partial a_{20}} = -(t_1 - a_{20}) = 0.309$$

1-4. Optimization

Backpropagation



$t_1 = 0.3$

$$E = \frac{1}{2}\sum_{i=1}^{2}(t_i - y_i)^2$$

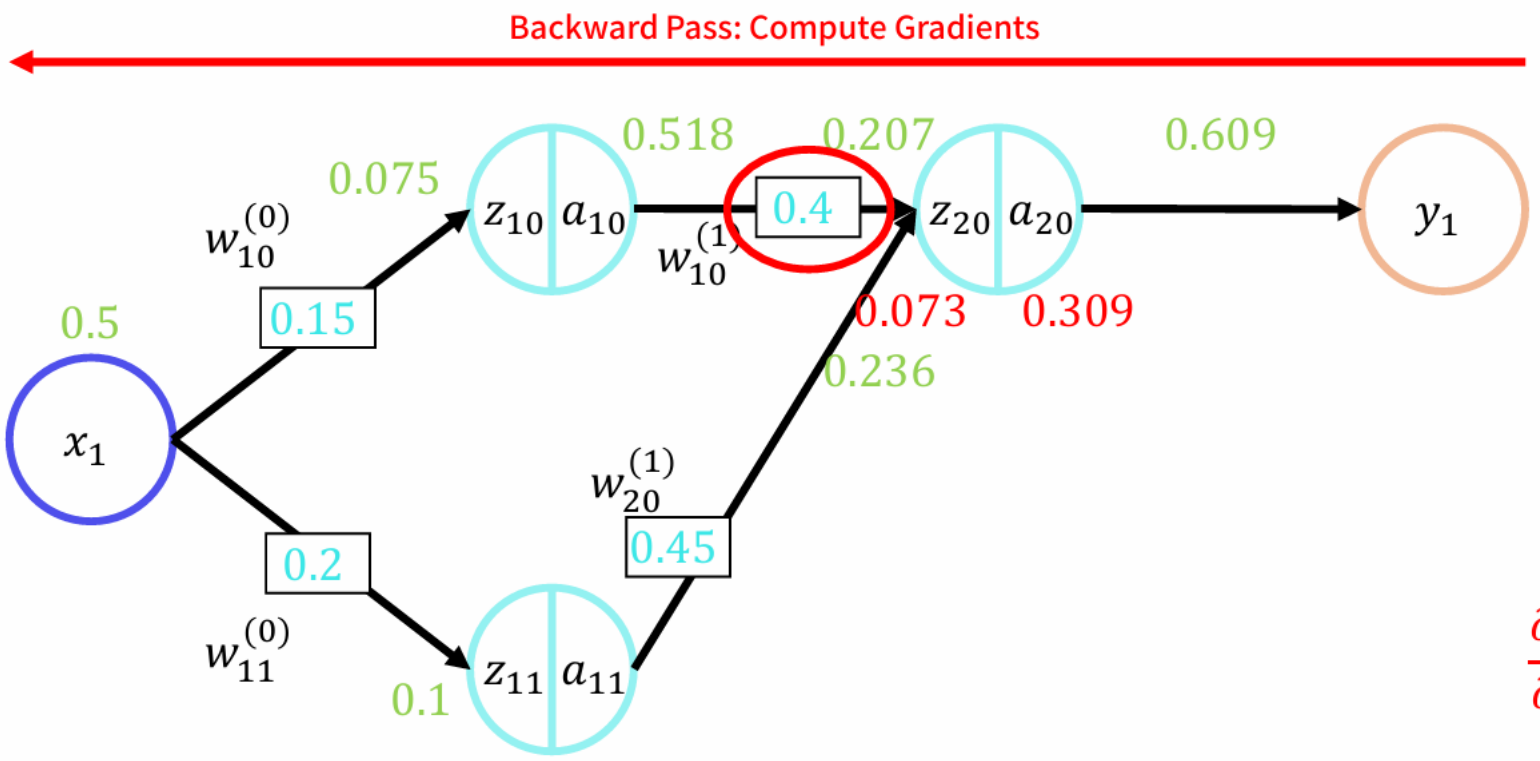
$$\frac{\partial E}{\partial w_{10}^{(1)}} = \frac{\partial E}{\partial a_{20}}\frac{\partial a_{20}}{\partial z_{20}}\frac{\partial z_{20}}{\partial w_{10}^{(1)}}$$

$$a_{20} = sigmoid(z_{20})$$

$$\frac{\partial a_{20}}{\partial z_{20}} = sigmoid(z_{20})\big(1 - sigmoid(z_{20})\big)$$
$$= 0.238$$

1-4. Optimization

Backpropagation



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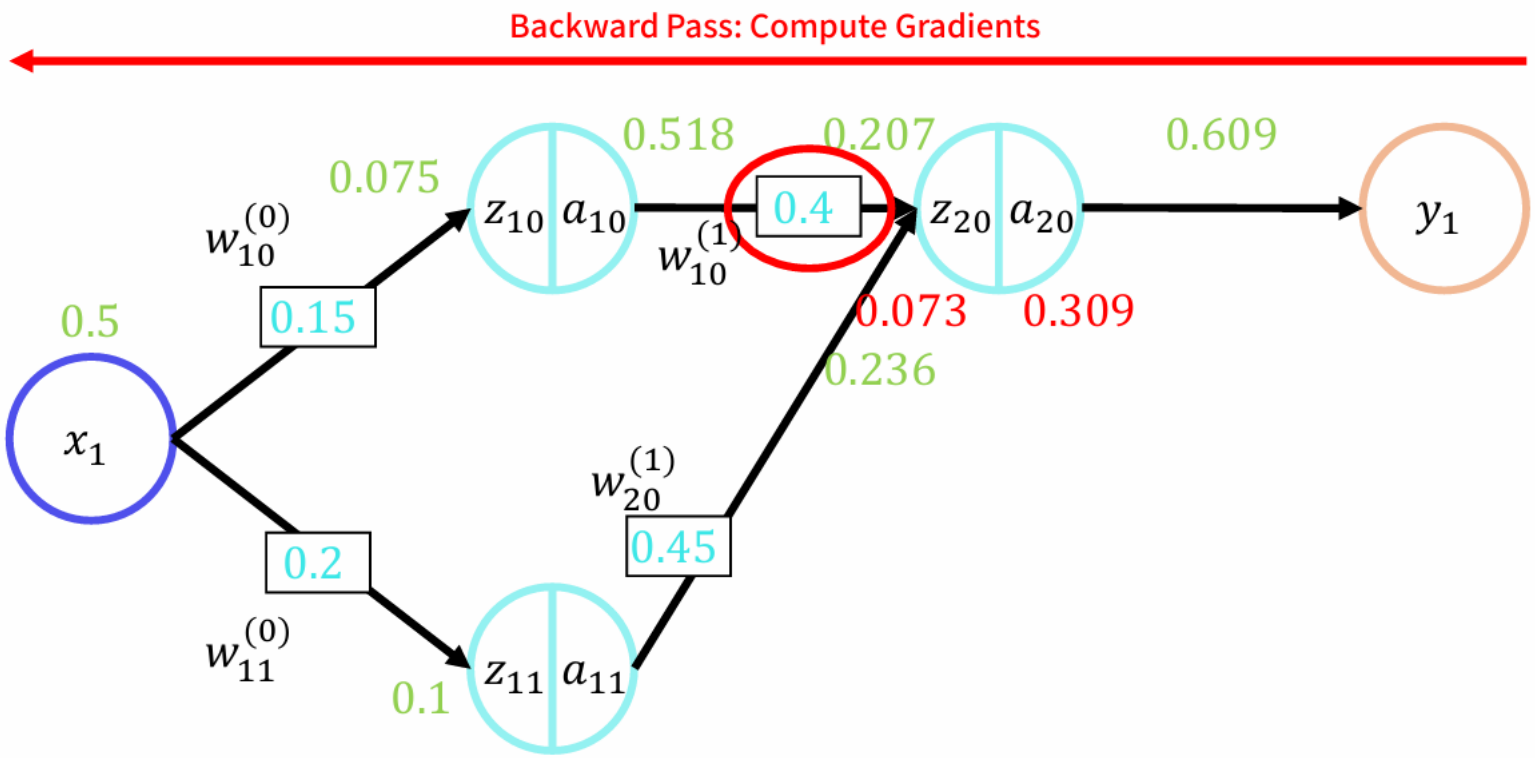
1-4. Optimization

Backpropagation



$t_1 = 0.3$

$$E = \frac{1}{2}\sum_{i=1}^{2}(t_i - y_i)^2$$

$$\frac{\partial E}{\partial w_{10}^{(1)}} = \frac{\partial E}{\partial a_{20}}\frac{\partial a_{20}}{\partial z_{20}}\frac{\partial z_{20}}{\partial w_{10}^{(1)}}$$

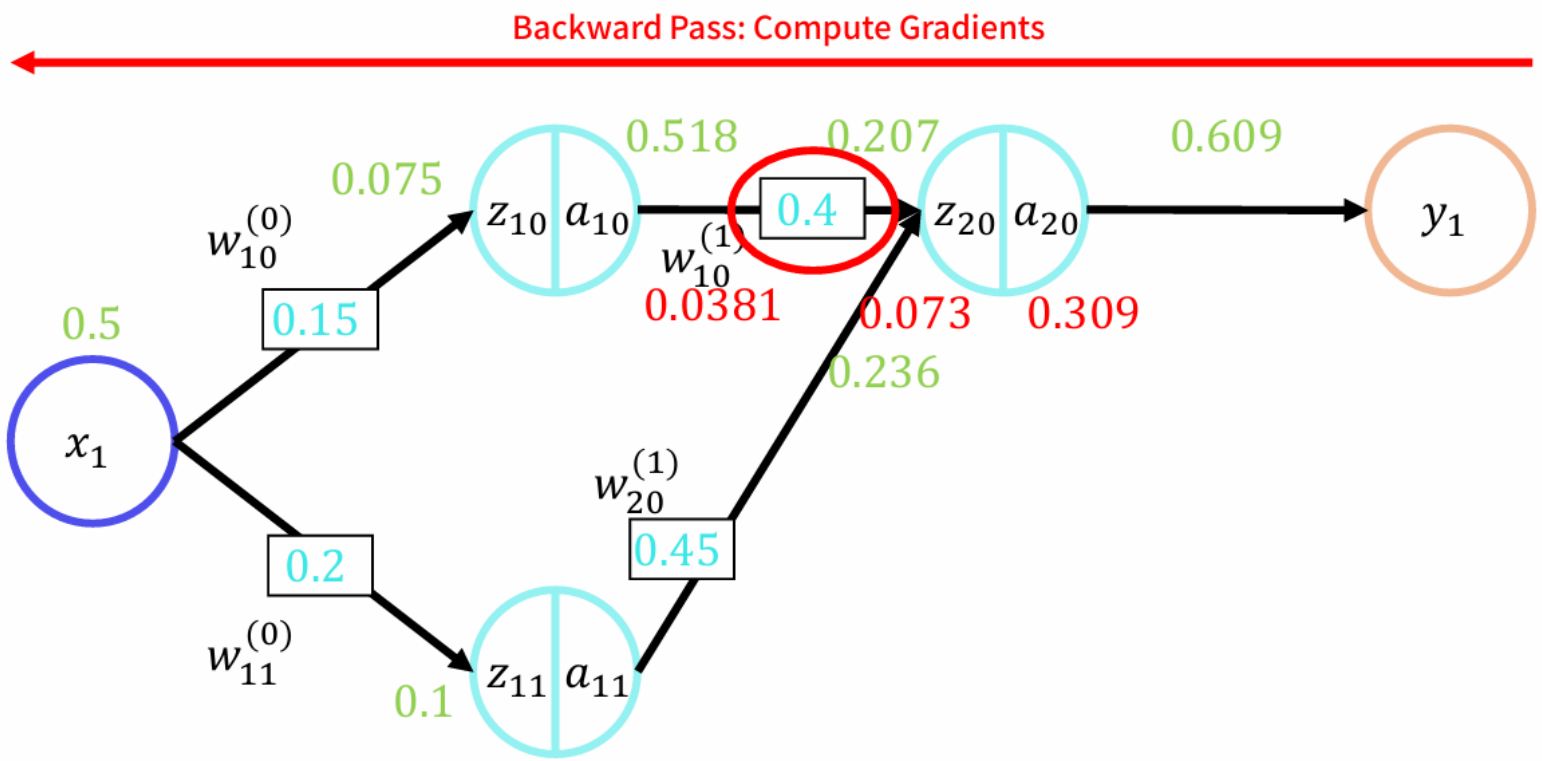
$$z_{20} = a_{10}w_{10} + a_{11}w_{11}$$

$$\frac{\partial z_{20}}{\partial w_{10}^{(1)}} = a_{10}$$

$$= 0.518$$

1-4. Optimization

Backpropagation



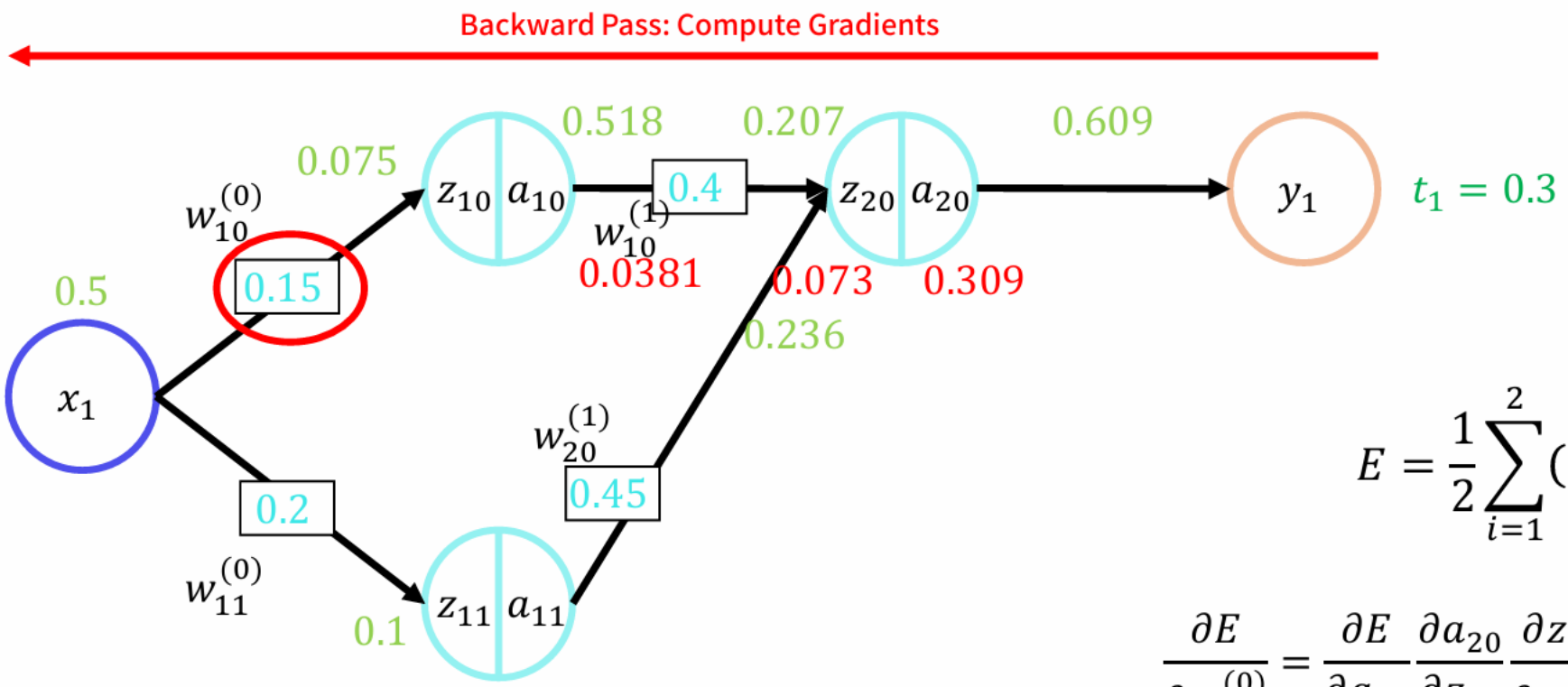
$$E = \frac{1}{2}\sum_{i=1}^{2}(t_i - y_i)^2$$

$$\frac{\partial E}{\partial w_{10}^{(1)}} = \frac{\partial E}{\partial a_{20}}\frac{\partial a_{20}}{\partial z_{20}}\frac{\partial z_{20}}{\partial w_{10}^{(1)}}$$

$$= 0.0381$$

1-4. Optimization

Backpropagation

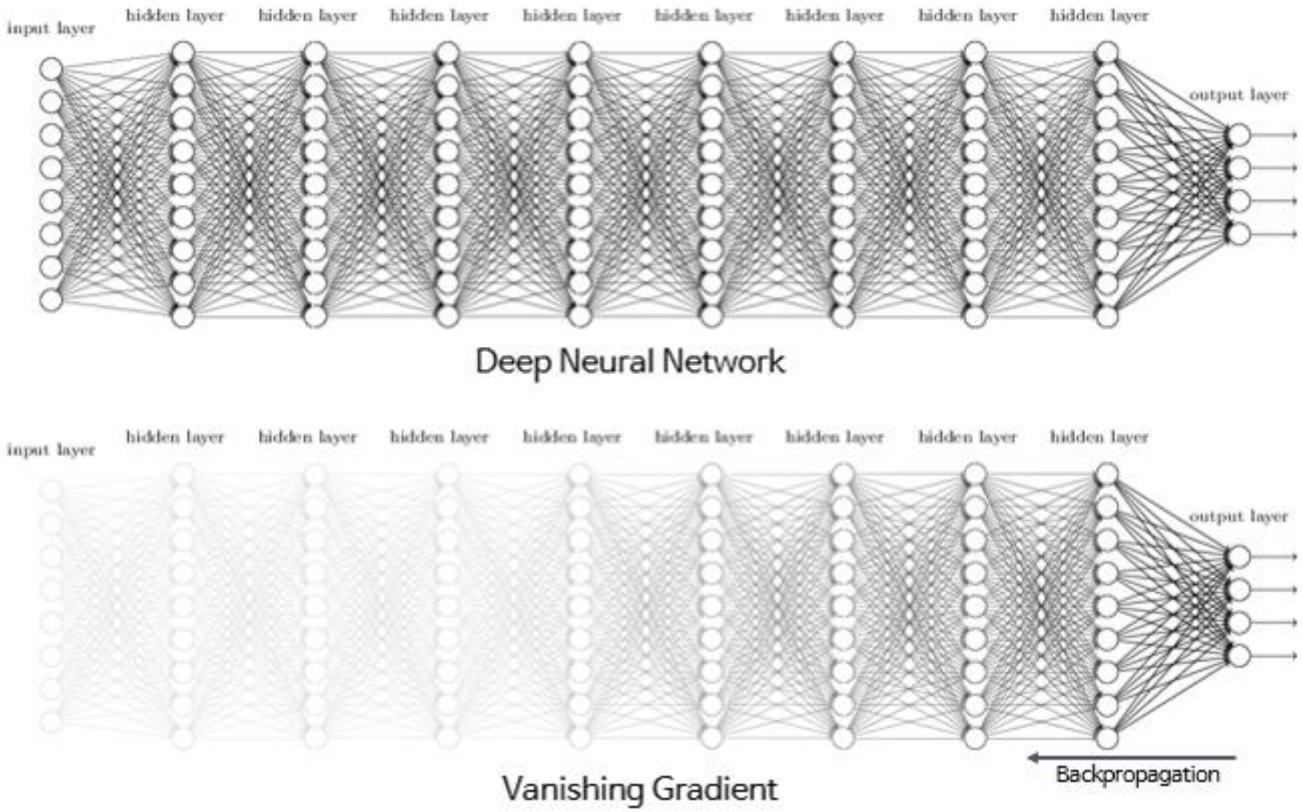


$$E = \frac{1}{2}\sum_{i=1}^{2}(t_i - y_i)^2$$

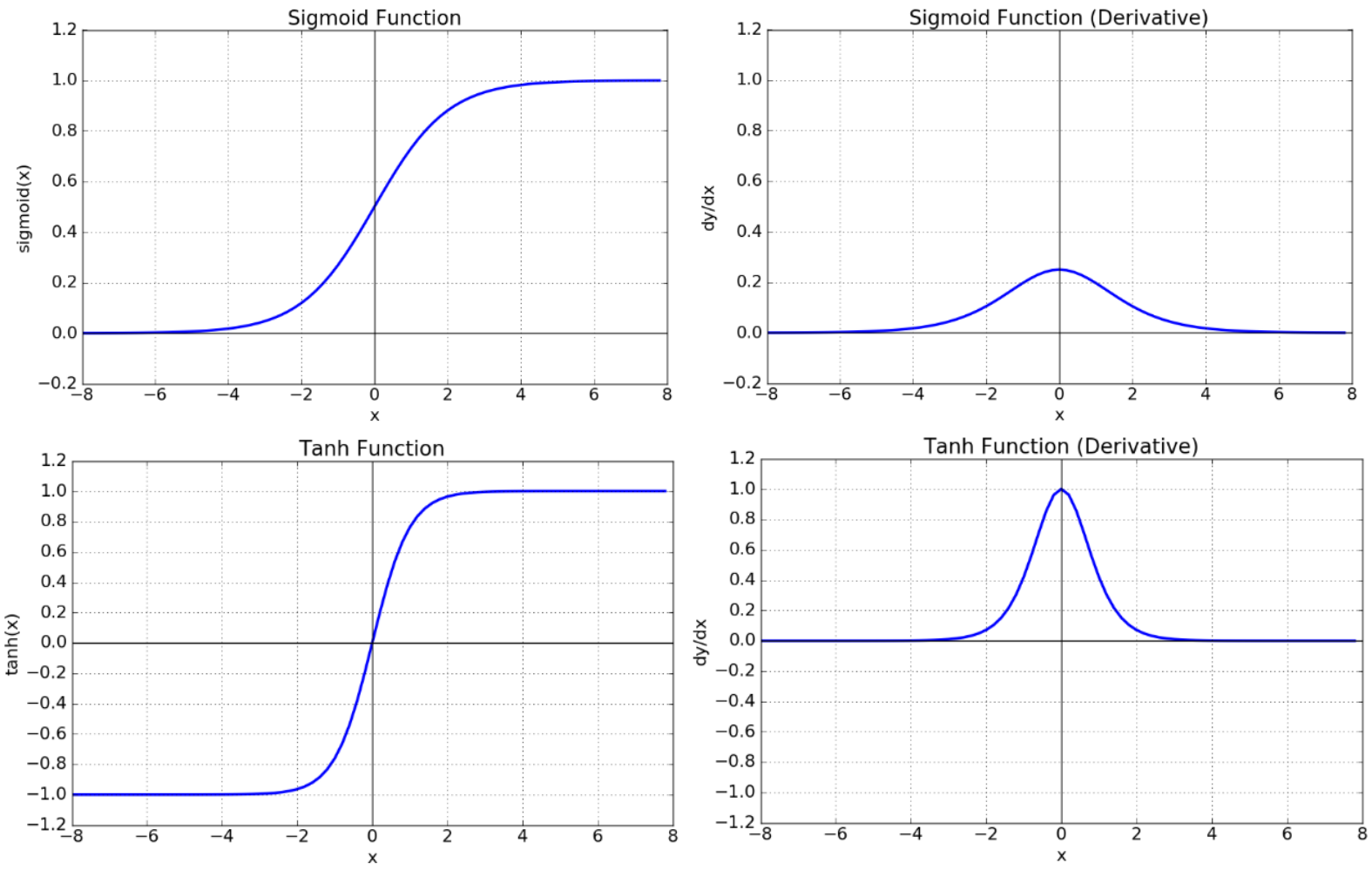
$$\frac{\partial E}{\partial w_{10}^{(0)}} = \frac{\partial E}{\partial a_{20}}\frac{\partial a_{20}}{\partial z_{20}}\frac{\partial z_{20}}{\partial w_{10}^{(1)}}\frac{\partial w_{10}^{(1)}}{\partial a_{10}}\frac{\partial a_{10}}{\partial z_{10}}\frac{\partial z_{10}}{\partial w_{10}^{(0)}}$$

1-5. Gradient Vanishing

as the network depth or sequence length increases, **<u>the gradient decrease</u>**, slowing the training process



1-5. Gradient Vanishing

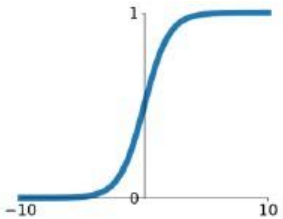


1-5. Gradient Vanishing

Activation Function (Non-Linearity)

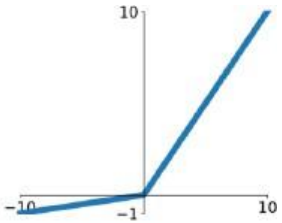
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



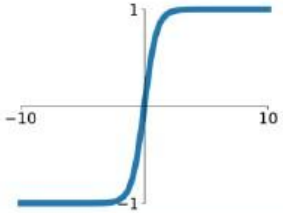
Leaky ReLU

$$\max(0.1x, x)$$



tanh

$$\tanh(x)$$

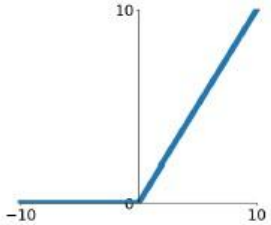


Maxout

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

ReLU

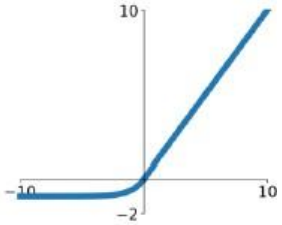
$$\max(0, x)$$



Dying ReLU?

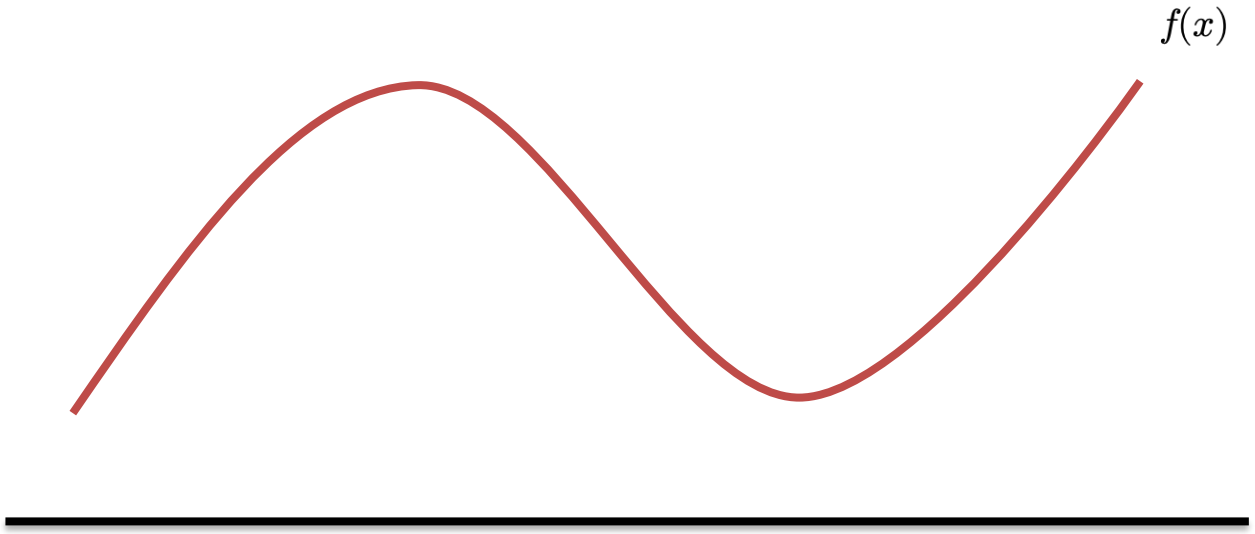
ELU

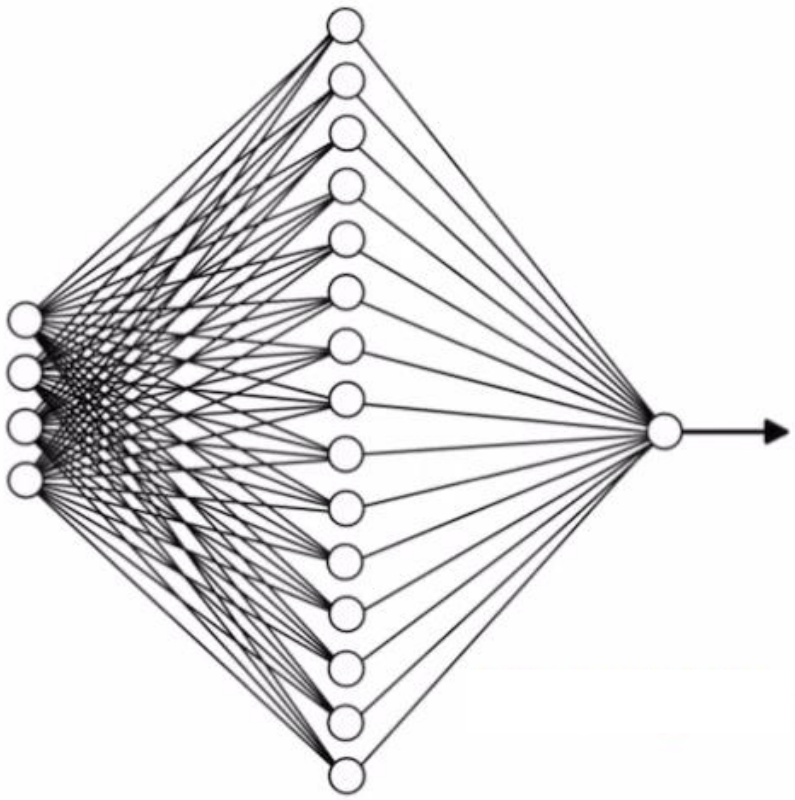
$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$



1-6. Universal Approximation Theorem

Upper Bound of Neural Network

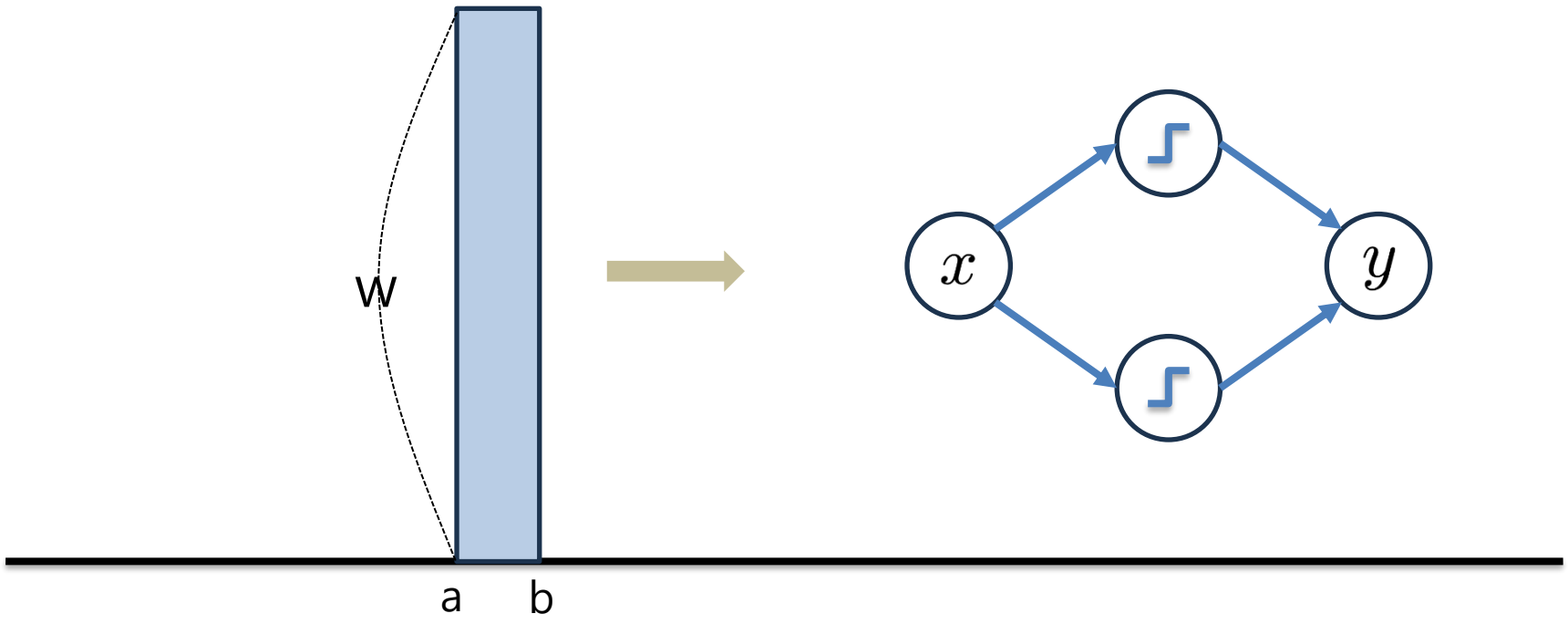




1-6. Universal Approximation Theorem

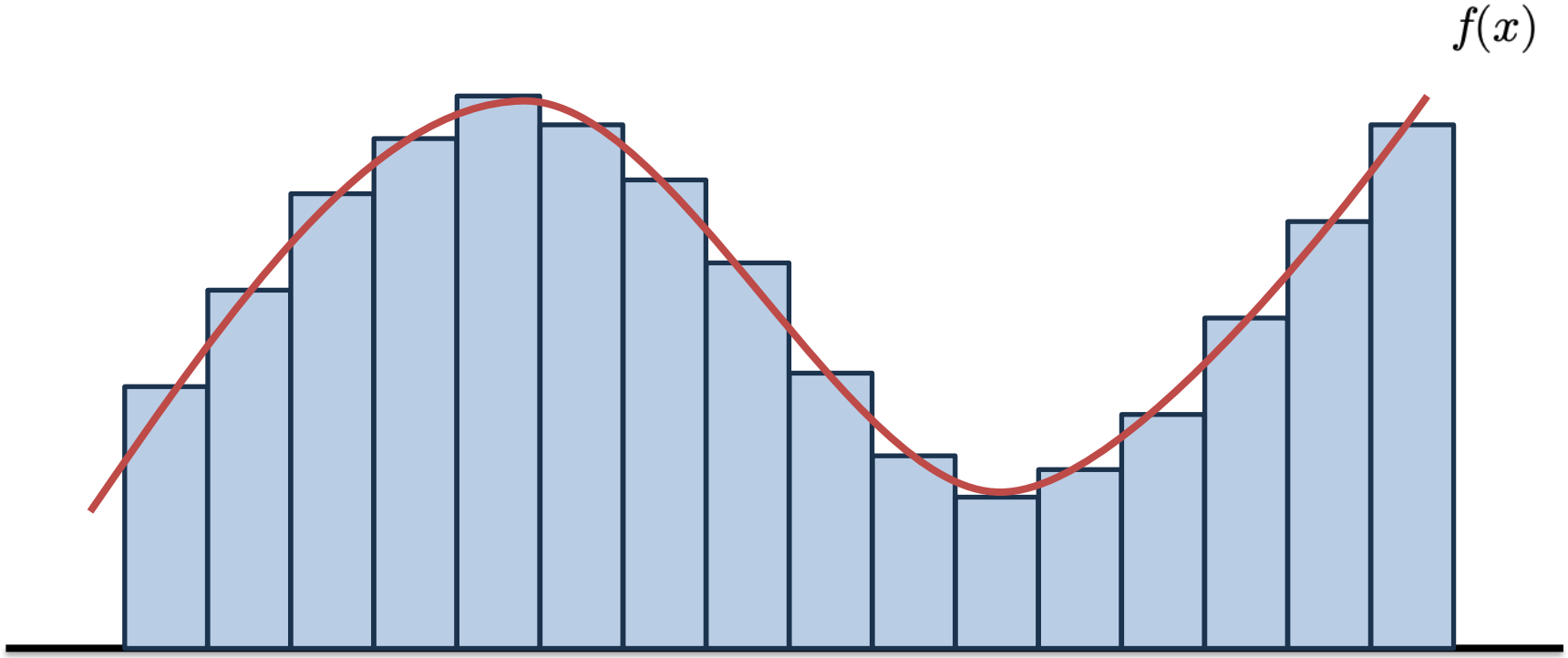
Universal Approximation Theorem

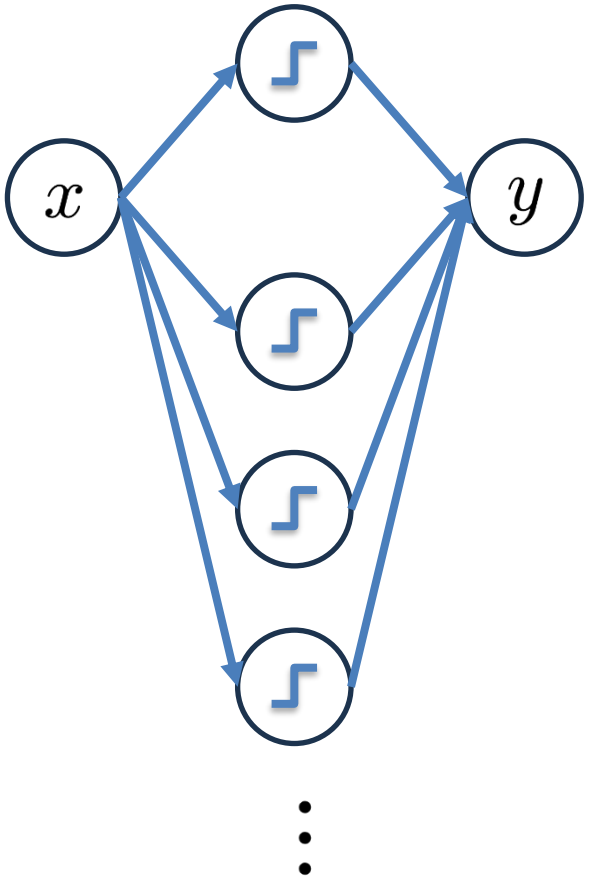
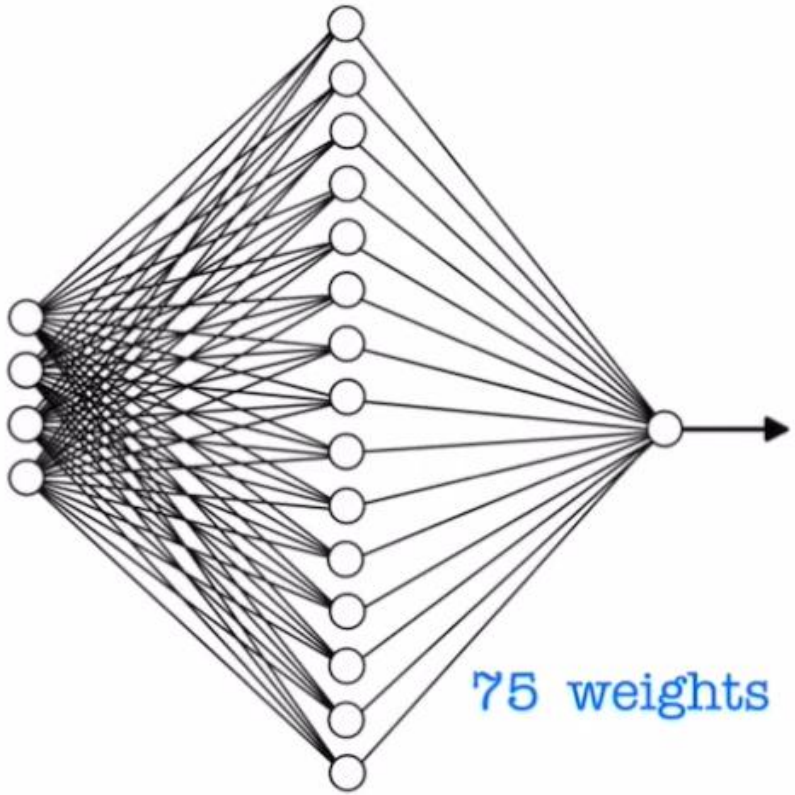
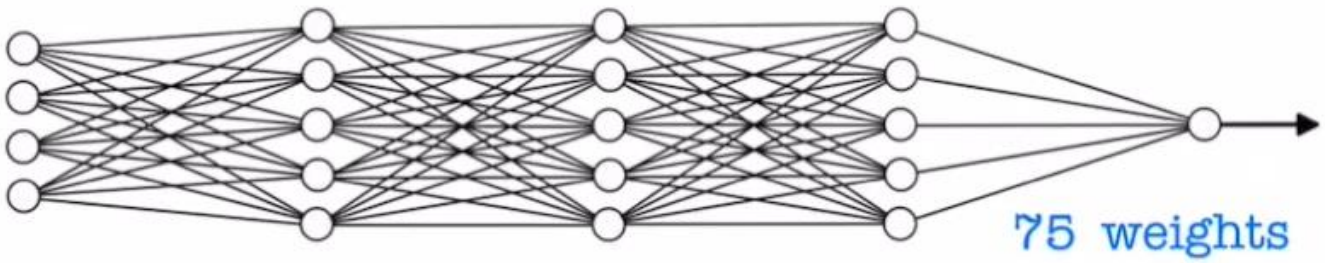
$$w \times (\delta(x-a) - \delta(x-b))$$



1-6. Universal Approximation Theorem

Universal Approximation Theorem





1-6. Universal Approximation Theorem